

SPARSE OBSERVATIONS AND PATTERN COMPLEXITY OF SAFE FEEDBACKS

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ABSTRACT. A safe state feedback generates a sequence of control instructions from each initial state. We study the growth of the number of patterns seen at selected times, with a fixed control period τ . The main question is whether one observation sequence detects complexity uniformly over all safe Borel feedbacks. We answer it when all legal blocks have a common endpoint map conjugate to a power of a mixing one-sided shift of finite type. The minimum rate is $\log r/\tau$, where r is the minimum number of block safety domains covering the state space, and every observation sequence with diverging gaps attains it. When controls have different successors, we bound the initial-state sets producing specified words by products of inverse-transfer operators. A contraction condition, combined with the finite covering lemma of Kerr and Li, determines the minimum maximal-pattern entropy; the observation times in this argument may depend on the feedback. A separate deletion argument gives common attaining sequences for countably many name languages. Symbolic and affine examples distinguish these conclusions from finite-horizon safe covering and from invariance entropy.

1. INTRODUCTION

Consider a discrete-time control system and a compact set Q in which the state must remain. A feedback chooses a control from the current state; the same rule is used at every subsequent step. Starting from different points of Q produces different sequences of controls. We ask how few patterns these sequences can have when they are observed at selected times. The feedback is allowed to be Borel, so the question includes rules with irregular decision regions.

A simple symbolic system illustrates the choice involved. Let $Q = \{0, 1, 2\}^{\mathbb{N}_0}$. At a state x , either control different from x_0 is safe, and both safe controls send x to the shift σx . A feedback is therefore a Borel function s satisfying $s(x) \neq x_0$. Its control sequence is

$$s(x), s(\sigma x), s(\sigma^2 x), \dots$$

The state orbit is already determined, but the observed labels depend on how s resolves the overlaps of the three safety sets. Choosing $s(x) = 1$ when $x_0 = 0$ and $s(x) = 0$ otherwise uses two labels and realizes every binary pattern at any finite set of times. Can another Borel choice reduce this complexity? If the observation times are sufficiently far apart, our closed-cover theorem says that it cannot. We return to this system in Sections 2.1 and 3.

More generally, fix a control period τ : the chosen instruction is a word of τ controls, all of whose intermediate states must stay in Q . Write $P_s(F)$ for the number of control-block patterns produced by a feedback s on a finite set F of block indices. For an increasing sequence $A = (a_i)$, its sequence entropy is

$$h_A(s) = \limsup_{n \rightarrow \infty} \frac{1}{n\tau} \log P_s(\{a_1, \dots, a_n\}).$$

The normalization counts observations, each charged the block length τ ; it does not count the intervening gaps. Maximizing over n -element windows before taking the limit gives the

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maximal-pattern entropy $h^*(s)$. Section 2 gives the definitions and the elementary limit argument.

There are two orders in which to choose a feedback and an observation sequence:

$$L_\tau = \sup_A \inf_s h_A(s), \quad M_\tau = \inf_s h^*(s).$$

For each fixed s , one has $h^*(s) = \max_A h_A(s)$ (Corollary 5.5). Thus M_τ allows the observation sequence to be chosen after the feedback. For L_τ , the sequence must be fixed first and give a lower bound for every safe feedback. The inequality $L_\tau \leq M_\tau$ is immediate; the opposite inequality requires a way to choose observation times in common.

Invariance entropy addresses a related question at consecutive times. It counts the smallest collection of control words that keeps every initial state safe for a finite horizon, and measures growth per elapsed time [7]. Such a collection can be chosen anew for each horizon. A feedback, by contrast, must make compatible choices whenever a trajectory returns to the same state. Sparse observations also retain information which a consecutive-time rate can miss. These differences matter even after optimization: the Thue–Morse realization in Section 6.5 has invariance entropy zero and $L_1 = M_1 = \log 2$. If the control period is allowed to vary, the infimum of the pattern rates returns to invariance entropy (Proposition 6.1).

We approach the fixed-period problem in two ways. First suppose that every legal block has the same endpoint map T , and that T is conjugate to a power of the shift on a mixing one-sided shift of finite type. Let r be the minimum number of block safety domains covering Q . Corollary 3.5 gives

$$\inf_s h_A(s) = L_\tau = M_\tau = \frac{\log r}{\tau} \quad \text{whenever } a_{i+1} - a_i \rightarrow \infty.$$

The upper bound uses an ordered minimum subcover. For the lower bound, any refining Borel partition has at least r nonmeager atoms. Each of these atoms occupies most of some cylinder in the sense of Baire category. Mixing joins sufficiently separated cylinders, and openness of the shift allows the resulting orbit to avoid the meager exceptions. This proves the bound on one prescribed sequence, simultaneously for all Borel feedbacks.

When the successor depends on the chosen control, this concatenation argument is no longer available. The second approach instead bounds the size of the sets of initial states producing specified control words. For a fixed list of $k - 1$ allowed labels at each time, a product of safe inverse-transfer operators bounds the mass carried by all words in those lists. If these bounds decay exponentially, the actual names cannot be covered by few such list boxes. The finite covering lemma of Kerr and Li then supplies k independent label choices on arbitrarily large sets of coordinates. Theorem 4.1 deduces $h^*(s) \geq \log k / \tau$ for every feedback, and $M_\tau = \log r / \tau$ when $k = r$. The independent positions may depend on s , which is exactly the obstacle to computing L_τ by the same proof.

The combinatorial input is due to Kerr and Li [8, Lemma 3.3]; it also follows from Huang and Ye’s sharper estimates [4, Lemma 4.1 and Remark 4.2]. The application here is the estimate for feedback fibers, with the safety restrictions retained at every intermediate time. The reference probability need not be invariant. Two examples distinguish the product condition from simpler tests: separated density peaks defeat the scalar test for a fixed measure, and a three-coordinate matrix product contracts even though the single-step density test fails for every full-support reference probability. Appendix C explains the exact matrix reduction and gives a general sufficient condition using a common positive left weight.

The closed-cover argument is related to Romagnoli’s comparison of an excluded-letter cover with its clopen refinements [10]. Here the refinements may be Borel, and a sparse sequence is prescribed before the refinement is chosen. The measurable common-sequence results of Huang, Maass, and Ye [3, Lemma 3.3 and Theorem 3.5] concern Shannon entropy relative to an invariant probability. Our counts include every realized pattern, even one supported on exceptional states. The broader sequence-entropy and independence framework is developed in [2, 5, 6, 8, 9].

The common-sequence question returns in Section 5: countably many name languages can share an attaining sequence even without shift invariance. Section 6 then explains why fixing the control period is essential, using a forced symbolic feedback and the Thue–Morse system. The affine examples in Section 7 compare two ways to obtain a lower bound: expansion makes individual fibers small, whereas protected inverse branches realize all sampled words. Their difference remains visible in a four-branch system for which M_1 is known but L_1 is not. The appendices supply the finite combinatorial estimate, the matrix criterion, the longer counts for this affine system, and a counterexample to pointwise common attainment. Each can be read with the result in the main text that calls for it.

2. SAFE FEEDBACKS AND THEIR OBSERVED NAMES

2.1. The control rule. Let X be a metrizable space and let $f_u : X \rightarrow X$ be continuous for each control $u \in U$. Initially U is finite; when we later consider invariant partitions we also allow an arbitrary control set. For a word $w = (w_0, \dots, w_{N-1})$, define

$$\varphi(0, x, w) = x, \quad \varphi(j+1, x, w) = f_{w_j}(\varphi(j, x, w)).$$

Throughout, all control sequences in $U^{\mathbb{N}_0}$ are admissible. Let $Q \subset X$ be nonempty, compact and *controlled invariant*: from every $x \in Q$ some infinite control sequence stays in Q . We use $\mathbb{N} = \{1, 2, \dots\}$, $\mathbb{N}_0 = \{0, 1, \dots\}$ and natural logarithms.

Fix $\tau \in \mathbb{N}$ and write $W = U^\tau$. The safety domain of a block w and its endpoint map are

$$K_w = \{x \in Q : \varphi(j, x, w) \in Q \ (0 \leq j \leq \tau)\}, \quad g_w(x) = \varphi(\tau, x, w). \quad (2.1)$$

Each K_w is closed. These sets cover Q , since a safe infinite control provides a safe first block. Their overlaps describe the choices available to a feedback.

A *safe feedback* is a Borel function $s : Q \rightarrow W$ such that $x \in K_{s(x)}$ for every $x \in Q$. Its block-boundary map is $T_s(x) = g_{s(x)}(x)$. This map is Borel and takes Q into itself. Applying the same rule at successive states keeps the entire trajectory in Q , including the times between block boundaries. We denote the class of all these rules by $\mathcal{S}_\tau$. The rule uses the current state only: there is no additional clock or memory variable. Continuity, or dependence on finitely many coordinates, is not required.

Sometimes we specify a finite block collection $W \subset U^\tau$ whose safety domains cover Q . In that case $\mathcal{S}_\tau$ and its optimization values are relative to that collection. Unless such a restriction is stated, $W = U^\tau$.

2.2. Counting at selected times. Starting at x , the rule produces the infinite name $(s(T_s^j x))_{j \geq 0}$. For a finite set $F \subset \mathbb{N}_0$ put

$$P_s(F) = |\{(s(T_s^j x))_{j \in F} : x \in Q\}|, \quad P_s(\emptyset) = 1.$$

Thus a pattern is counted if it occurs on at least one complete safe trajectory. Write $\mathcal{S}$ for the strictly increasing infinite sequences in $\mathbb{N}_0$. For $A = (a_i) \in \mathcal{S}$, set

$$h_A(s) = \limsup_{n \rightarrow \infty} \frac{\log P_s(\{a_1, \dots, a_n\})}{n\tau}, \quad p_s^*(n) = \max_{|F|=n} P_s(F),$$

and

$$h^*(s) = \lim_{n \rightarrow \infty} \frac{\log p_s^*(n)}{n\tau}.$$

The maximum exists because the possible counts are integers between 1 and $|W|^n$. Splitting a window into parts of sizes m, n gives $p_s^*(m+n) \leq p_s^*(m)p_s^*(n)$, so Fekete's lemma gives the last limit. In particular $h_A(s) \leq h^*(s)$.

The index a_i refers to the a_i th block, observed at physical time τa_i . For example, when $\tau = 2$ and $A = (0, 3, 6, \dots)$, the observations are made at times $0, 6, 12, \dots$, but the denominator for n observations is $2n$. This convention measures the diversity per observed instruction, with the block length accounted for.

The two optimized quantities are

$$L_\tau = \sup_{A \in \mathcal{J}} \inf_{s \in \mathcal{S}_\tau} h_A(s), \quad M_\tau = \inf_{s \in \mathcal{S}_\tau} h^*(s). \quad (2.2)$$

They satisfy $L_\tau \leq M_\tau$. If r safety domains cover Q , selecting the first legal block in that ordered subcover is a Borel feedback using at most r labels. It has at most r^n patterns on every n -element window. Hence

$$L_\tau \leq M_\tau \leq \frac{\log r}{\tau}.$$

Our main results give conditions for this elementary upper bound to be sharp. The infimum defining M_τ is always attained: maximal-pattern rates have the form $\log m/\tau$ for integers m (Appendix A).

In the three-symbol system from the introduction, the legal domains are $K_u = \{x : x_0 \neq u\}$ and $T_s = \sigma$ for every feedback. Illegal controls may be sent to an isolated fixed point $\dagger$ outside Q ; the resulting maps on $Q \sqcup \{\dagger\}$ are continuous. Two domains cover Q . The rule $s_0(x) = 1$ on $[0]$ and $s_0(x) = 0$ elsewhere has $P_{s_0}(F) = 2^{|F|}$: prescribe $x_j = 0$ or 1 to obtain, respectively, label 1 or 0 at each $j \in F$. This verifies the upper value $\log 2$ directly. The harder question is a lower bound valid for every Borel choice. The next section proves it on every sequence with diverging gaps. Here $[v]$ denotes the cylinder of sequences beginning with a word v ; we use this notation in all subsequent shift examples.

2.3. Observing a partition instead of a control. A finite Borel partition may split the domain assigned to one block into several observed atoms. Assigning a safe block to each atom gives an invariant cover in the sense of [7, Definition 2.8], with the underlying cover required to be a partition. We call this an *invariant partition*. Its state map still applies the assigned block, but its atom names may distinguish trajectories that have the same control name. The following merging argument shows why this additional distinction does not change the infima.

Lemma 2.1. *The infima in (2.2), and the inner infimum for each fixed A , are unchanged if finite Borel invariant partitions are allowed and atom names are counted.*

Proof. Merge the atoms assigned the same block. Their union is Borel and lies in that block's safety domain, so the merged partition defines the same state map. Projection of atom labels to block labels is onto every realized finite pattern set. It can only decrease counts. Conversely, the nonempty fibers of a selector are a finite invariant partition whose atom and block counts agree. Taking the two infima proves the assertion. $\square$

Example 2.2 (Observation labels and physical controls). Let $Q = \{0, 1\}^{\mathbb{N}_0} \cup \{2^\infty\}$. Control $u = x_0$ sends x to σx and every other control sends it to an isolated cemetery point. Safety forces $s(x) = x_0$. The n -position pattern count is $2^n + 1$ for $n \geq 1$. Renaming the observed label 2 as 0 reduces it to 2^n , with the same exponential rate. Yet no one physical control is safe on both $[0]$ and $\{2^\infty\}$. Such a renaming is an observation factor, not a merged safe feedback. Lemma 2.1 merges only atoms assigned the same block.

2.4. Finite-horizon covering. For later comparison, let $r_{\text{inv}}(N, Q)$ be the least size of a finite set $S \subset U^N$ such that from every $x \in Q$ at least one word in S stays safe through time N . Set this number to $+\infty$ if no finite spanning family exists. Invariance entropy is

$$h_{\text{inv}}(Q) = \limsup_{N \rightarrow \infty} \frac{\log r_{\text{inv}}(N, Q)}{N}. \quad (2.3)$$

This is the discrete-time definition in [7, Definition 2.2], with natural logarithms and the full control space $U^{\mathbb{N}_0}$. The words actually produced by a feedback form a spanning family. Conversely, a spanning family by itself prescribes no consistent choice at later states. We will use this distinction both in computing examples and in comparing the two entropy notions in Section 6.

3. COMMON SUCCESSORS AND A CLOSED-COVER FORMULA

Suppose that every legal control block sends a state to the same successor $T(x)$. A feedback then chooses labels along a fixed orbit, just as in the excluded-letter example. Its label fibers form a Borel partition refining the safety cover. We first solve this partition problem on a mixing shift of finite type. The control conclusion will follow by applying the formula to the block safety domains.

We use the counting convention of Section 2 for a finite Borel partition α and a Borel map $T : Q \rightarrow Q$ on a nonempty compact metric space. In this notation

$$P_\alpha(F) = \# \bigvee_{j \in F} T^{-j} \alpha$$

counts nonempty atoms, or equivalently the realized α -names on F . In the present partition problem the scale is one:

$$h_A(T, \alpha) = \limsup_n \frac{1}{n} \log P_\alpha(\{a_1, \dots, a_n\}), \quad (3.1)$$

$$p_\alpha^*(n) = \max_{|F|=n} P_\alpha(F), \quad h^*(T, \alpha) = \lim_n \frac{1}{n} \log p_\alpha^*(n). \quad (3.2)$$

As before, submultiplicativity gives the last limit and $h_A(T, \alpha) \leq h^*(T, \alpha)$. We restore the factor $1/\tau$ when applying the result to control blocks.

Let $\mathcal{K} = (K_w)_{w \in W}$ be a finite closed cover of Q , and write

$$r = N(\mathcal{K}) = \min\{|I| : I \subset W, \bigcup_{w \in I} K_w = Q\}.$$

The set $\mathcal{B}(\mathcal{K})$ consists of all finite Borel partitions refining $\mathcal{K}$: each atom must be contained in at least one K_w . Equivalently, one may first consider all Borel maps $s : Q \rightarrow W$ with $x \in K_{s(x)}$, and then allow arbitrary finite Borel refinements of their fibers.

Let H be a primitive finite 0–1 matrix, with vertex set V , and let

$$Y_H = \{y \in V^{\mathbb{N}_0} : H_{y_j y_{j+1}} = 1 \text{ for all } j \geq 0\}.$$

Primitivity means that all sufficiently large powers of H have strictly positive entries. Write σ for the one-sided shift.

Theorem 3.1 (Closed covers of mixing shifts). *Suppose $T : Q \rightarrow Q$ is topologically conjugate to $\sigma^p : Y_H \rightarrow Y_H$ for some primitive H and integer $p \geq 1$. Let $\mathcal{K}$ be any finite closed cover of Q . For every strictly increasing $A = (a_i)$ satisfying*

$$a_{i+1} - a_i \longrightarrow \infty, \quad (3.3)$$

one has

$$\min_{\alpha \in \mathcal{B}(\mathcal{K})} h_A(T, \alpha) = \min_{\alpha \in \mathcal{B}(\mathcal{K})} h^*(T, \alpha) = \log r. \quad (3.4)$$

An ordered difference partition of any minimum subcover attains both minima, simultaneously for every A satisfying (3.3). Consequently,

$$\max_A \inf_{\alpha \in \mathcal{B}(\mathcal{K})} h_A(T, \alpha) = \log r, \quad (3.5)$$

where the maximum ranges over all strictly increasing infinite sequences and is attained, for example, by $a_i = (i - 1)^2$.

The lower bound has two ingredients. Closedness of the cover forces at least r atoms of any refining partition to be nonmeager. Mixing then lets us prescribe any string of these atoms at sufficiently separated times. The separation threshold depends on the partition, but every sequence with diverging gaps eventually exceeds it. This is why one A can be fixed before choosing the partition.

3.1. A category and concatenation lemma. All category statements are relative to Q . A nonmeager Borel set is comeager in some nonempty open set, since Borel sets have the Baire property. We will replace each of several nonmeager atoms by a cylinder on which it is comeager, join the cylinders, and then remove the bad sets. For that last step we need openness: if F is continuous and open, the preimage of a closed nowhere dense set has empty interior. Consequently preimages of meager sets are meager.

Lemma 3.2. *Under the dynamical assumptions of Theorem 3.1, T is continuous, open, and onto. Given finitely many nonempty cylinder sets $C_1, \dots, C_q$ in symbolic coordinates, there is an integer $b \geq 1$ such that, for every $k \geq 1$, every $t_1, \dots, t_k \in \{1, \dots, q\}$, and all $0 \leq n_1 < \dots < n_k$ with $n_{j+1} - n_j \geq b$,*

$$\bigcap_{j=1}^k T^{-n_j} C_{t_j}$$

is nonempty and open.

Proof. Work on Y_H through the conjugacy. The image under σ of a prefix cylinder of length at least two is the cylinder obtained by removing its first symbol. The image of a one-symbol cylinder is the union of the permitted successor cylinders. Hence σ is open. Every vertex has a predecessor, so every infinite path has a predecessor; thus σ is onto. These properties pass to powers and conjugates.

Choose within each C_j a nonempty prefix cylinder, and extend their defining words, if necessary, to a common length ℓ . Choose N with $H^n > 0$ for every $n \geq N$, and then choose b with $pb - \ell + 1 \geq N$. Place the word for C_{t_j} at coordinate pn_j . Between the end of one placed word and the start of the next there are $p(n_{j+1} - n_j) - \ell + 1$ edges to fill. Primitivity supplies a path of that length between any two endpoint vertices. Predecessors fill the finite segment before the first placed word, and successors provide an infinite continuation after the last. This constructs an actual point in the intersection. The intersection is open because it is a finite intersection of preimages of open sets under continuous maps. $\square$

Lemma 3.3. *Under the same assumptions, let α be a finite Borel partition with q nonmeager atoms. For every A satisfying (3.3) there exists $m \geq 1$ such that*

$$P_\alpha(\{a_1, \dots, a_n\}) \geq q^{n-m+1} \quad (n \geq m). \quad (3.6)$$

In particular $h_A(T, \alpha) \geq \log q$.

Proof. We first choose cylinders in which the selected atoms occupy a comeager set. We then join those cylinders along the prescribed times; the discarded subsets remain meager under pullback.

List the nonmeager atoms as $D_1, \dots, D_q$. The Baire property provides a nonempty cylinder C_j for which $C_j \setminus D_j$ is meager. Let b be supplied by Lemma 3.2, and choose m so that $a_{i+1} - a_i \geq b$ for all $i \geq m$.

Fix $n \geq m$ and any target labels $t_m, \dots, t_n$. The set

$$G = \bigcap_{i=m}^n T^{-a_i} C_{t_i}$$

is nonempty and open. Each power T^{a_i} is continuous and open, so

$$\bigcup_{i=m}^n T^{-a_i} (C_{t_i} \setminus D_{t_i})$$

is meager and cannot cover G . A point outside this union realizes all the specified atoms at the sampled times. Distinct target label strings give distinct patterns; restriction from the full window to this tail is onto. This proves (3.6). Dividing its logarithm by n and taking a limit inferior proves the asserted bound. $\square$

3.2. Proof of the exact formula.

Proof of Theorem 3.1. Let $\alpha \in \mathcal{B}(\mathcal{K})$. Assign to each atom D one index $w(D)$ with $D \subset K_{w(D)}$. Let J be the set of indices assigned to the nonmeager atoms. If the closed sets $(K_w)_{w \in J}$ did not cover Q , their complement would be a nonempty open set covered by the finitely many meager atoms of α , which is impossible. Thus $|J| \geq r$, and the number q of nonmeager atoms is at least r . Lemma 3.3 gives

$$h_A(T, \alpha) \geq \log q \geq \log r$$

for every A satisfying (3.3). Closedness of the cover members is used precisely in this passage from category to a full cover.

Choose a minimum subcover $K_{w_1}, \dots, K_{w_r}$, and set

$$E_j = K_{w_j} \setminus \bigcup_{k < j} K_{w_k}, \quad \alpha_0 = \{E_1, \dots, E_r\}. \quad (3.7)$$

Every E_j is Borel and is nonempty, since a minimum subcover has no redundant member. The partition refines $\mathcal{K}$ and has r atoms. Hence $\mathbf{P}_{\alpha_0}(F) \leq r^{|F|}$ for all finite F , giving $h_A(T, \alpha_0) \leq h^*(T, \alpha_0) \leq \log r$ for every A . The established lower bound and $h^* \geq h_A$ yield (3.4) and simultaneous attainment. For arbitrary A , the partition α_0 bounds the inner infimum in (3.5) by $\log r$. Any A satisfying (3.3) gives the reverse bound and attains the maximum. $\square$

3.3. An excluded-letter cover. For any finite cover $\mathcal{K}$ define its combinatorial sequence entropy by

$$h_A(T, \mathcal{K}) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log N \left(\bigvee_{i=1}^n T^{-a_i} \mathcal{K} \right).$$

For a partition this is (3.1). For overlapping sets, covering each finite orbit window need not correspond to one fixed refining partition used at every time.

Corollary 3.4. *Let $q \geq 3$, $Q = \{0, \dots, q-1\}^{\mathbb{N}_0}$, $T = \sigma$, and*

$$K_j = \{x : x_0 \neq j\} \quad (0 \leq j < q), \quad \mathcal{K} = (K_j)_{j=0}^{q-1}.$$

For every increasing infinite A ,

$$h_A(\sigma, \mathcal{K}) = \log \frac{q}{q-1}. \quad (3.8)$$

For every such A with diverging successive gaps,

$$\min_{\alpha \in \mathcal{B}(\mathcal{K})} h_A(\sigma, \alpha) = \log 2 > h_A(\sigma, \mathcal{K}). \quad (3.9)$$

Proof. The cover has no single member equal to Q , and any two distinct members cover Q . Thus $r = 2$, and (3.9) follows from Theorem 3.1 once (3.8) is proved.

For a window F of size n , the full shift has all q^n possible input vectors. Each member of the joined cover specifies one excluded letter at each coordinate, and covers $(q-1)^n$ of these vectors. Consequently the minimum number $c(n)$ of such boxes is at least $(q/(q-1))^n$. For completeness, the following elementary probabilistic covering bound gives the matching exponent. Choose R excluded-letter vectors independently and uniformly from the q^n possibilities. A fixed input vector is left uncovered with probability at most

$$\left(1 - \left(\frac{q-1}{q} \right)^n \right)^R \leq \exp \left(-R \left(\frac{q-1}{q} \right)^n \right).$$

Taking $R = \lceil (q/(q-1))^n (n \log q + 1) \rceil$ makes the sum of these probabilities over all input vectors smaller than one. Some choice therefore covers every vector. We have proved

$$\left(\frac{q}{q-1} \right)^n \leq c(n) \leq \left\lceil \left(\frac{q}{q-1} \right)^n (n \log q + 1) \right\rceil.$$

The count depends only on n , not on the positions of the sampled coordinates, so its exponential rate proves (3.8). $\square$

The excluded-letter cover and its covering exponent are already present in Romagnoli [10]; the calculation above is included to identify the sparse-time quantity without an external dependency. The consequence of Theorem 3.1 is the exact minimum over *all Borel* refinements for a sequence fixed in advance. It does not assert that every Borel refinement has consecutive-time entropy at least $\log 2$, or that every sampling sequence gives that lower bound.

3.4. Application to safe feedbacks. For a fixed period, the common-successor hypothesis identifies every feedback orbit with the same dynamical system. The preceding cover formula therefore applies to the control-label fibers.

Corollary 3.5 (Safe feedbacks with a common successor). *In the setting of Section 2.1, assume that there is a map $T : Q \rightarrow Q$ such that*

$$g_w(x) = T(x) \quad \text{for every } w \in W \text{ and every } x \in K_w, \quad (3.10)$$

and that T is conjugate to σ^p on a mixing one-sided shift of finite type. With $r = N((K_w)_{w \in W})$, for every A satisfying (3.3) one has

$$\min_{s \in S_\tau} h_A(s) = \min_{s \in S_\tau} h^*(s) = L_\tau = M_\tau = \frac{\log r}{\tau}. \quad (3.11)$$

The maximum defining L_τ is attained by every such A , and an ordered minimum-subcover selector attains all the indicated minima. The same minimum and minimax values hold when all finite Borel safe partitions, including every finite Borel refinement, are allowed. Every refinement satisfies the stated lower bound; no upper bound for each individual refinement is asserted.

Proof. For every legal selector, (3.10) gives $T_s = T$ pointwise. Its fibers form a finite Borel refinement of $(K_w)_{w \in W}$. Theorem 3.1, with normalization by τ , gives the lower bound at every A with diverging gaps. If $w_1, \dots, w_r$ form a minimum subcover, define $s_0(x) = w_j$ on the sets E_j of (3.7). This is Borel and legal at every point. Each selected block stays in Q at its intermediate times by (2.1), and the endpoint again lies in Q , so recursion gives an infinite safe closed-loop orbit for every initial state. The selector uses only r blocks and thus gives the upper bound for every window and every sampling sequence. This proves attainment, (3.11), and the outer maximum. Lemma 2.1 proves the last assertion. $\square$

At period one this gives $L_1 = M_1 = \log 2$ for the excluded-letter system from the introduction. At longer periods, the cover cardinality must be computed for whole blocks, as the following example shows.

3.4.1. An overlapping example at period two. Take

$$Q = \{0, 1, 2\}^{\mathbb{N}_0}, \quad X = Q \sqcup \{\dagger\}, \quad U = \{0, 1, 2\}, \quad \tau = 2,$$

where the isolated point $\dagger$ lies outside Q . Define

$$f_u(x) = \begin{cases} \sigma x, & x \in Q, \ x_0 \neq u, \\ \dagger, & x \in Q, \ x_0 = u, \end{cases} \quad f_u(\dagger) = \dagger.$$

Each f_u is continuous since the cases are clopen. For $w = (a, b)$,

$$K_{(a,b)} = \{x : x_0 \neq a, \ x_1 \neq b\}, \quad g_{(a,b)}|_{K_{(a,b)}} = \sigma^2.$$

All nine control blocks are available to Borel feedbacks, and each state permits four of them. A safety domain covers four of the nine two-symbol input blocks, so two domains cannot

cover Q . The three diagonal blocks $(0, 0), (1, 1), (2, 2)$ do cover Q , since some symbol differs from both x_0 and x_1 . Hence $r = 3$, and

$$L_2 = M_2 = \frac{\log 3}{2}.$$

An attaining selector is

$$c(x) = \min(\{0, 1, 2\} \setminus \{x_0, x_1\}), \quad s_0(x) = (c(x), c(x)).$$

It is clopen and legal everywhere. Its actual block-boundary states are $\sigma^{2k}x$. To prescribe any sequence of diagonal labels, use successive input blocks 11, 00, and 01 for the desired labels 0, 1, and 2, respectively. Therefore its actual name set is the full shift on these three blocks. The lower bound, however, holds for every legal Borel rule using any of the nine blocks.

The common object here is the state orbit; there need not be a nonconstant label observation shared by all feedbacks. To see this, suppose a map $\rho : U^2 \rightarrow V'$ and an observation $F : Q \rightarrow V'$ satisfy $\rho(s(x)) = F(x)$ for every legal Borel selector and every x . Any two blocks (a, b) and (a', b') are both legal at a state beginning with ij , where $i \notin \{a, a'\}$ and $j \notin \{b, b'\}$. A legal baseline selector may be changed at this single state to use either block, while remaining Borel and legal. Hence their ρ values agree. Every pair of blocks is legal together somewhere, so ρ is constant. Thus the argument uses refinements of an overlapping cover, rather than a label forced at each state.

3.5. Why the successor map matters. Changing the successor can prevent a region from being visited again. The minimum safety cover still counts that region, but its label may contribute only to the initial part of a name. The next example has $r = 3$ and $L_1 = M_1 = \log 2$. Thus continuity and openness of the individual branches do not replace the common-successor assumption.

Let $Q = \{0, 1\}^{\mathbb{N}_0}$, let $\mathcal{C} = \{0, 10, 11\}$ be a set of finite words, and take $U = \mathcal{C} \times \{0, 1\}$ and $\tau = 1$. For $e \in \{0, 1\}$ let P_e add e modulo two to every coordinate. On $X = Q \sqcup \{\dagger\}$ define

$$f_{(c,e)}(x) = \begin{cases} P_e \sigma x, & x \in [c], \\ \dagger, & x \in Q \setminus [c], \end{cases} \quad f_{(c,e)}(\dagger) = \dagger. \quad (3.12)$$

The cylinders $[0], [10], [11]$ are disjoint and cover Q . Each state has exactly two legal controls with different successors. The global maps $f_{(c,e)}$ are continuous, and their legal restrictions are open. The safety domains are $K_{(c,e)} = [c]$, and $r = 3$.

Proposition 3.6. *For the system (3.12), over all legal Borel feedbacks and arbitrary Borel mixtures,*

$$\min_s h_{A_0}(s) = \min_s h^*(s) = \max_A \inf_s h_A(s) = \log 2 < \log r, \quad A_0 = (0, 1, 2, \dots). \quad (3.13)$$

There is one clopen legal feedback s_f satisfying $h_A(s_f) = \log 2$ for every increasing infinite A . The same full-class values hold if arbitrary finite Borel atom refinements are allowed.

Proof. For the upper bound we choose the flip that makes the next first coordinate zero. After that first step only two labels remain. For the lower bound we recover the input prefix from any actual control word, whatever feedback produced it.

Let $c(x)$ be the unique member of $\mathcal{C}$ with $x \in [c(x)]$, and set

$$s_f(x) = (c(x), x_1), \quad T_f x = P_{x_1} \sigma x.$$

This is a clopen selector, legal at every state. Its four nonempty initial control fibers, for labels $(0, 0), (0, 1), (10, 0), (11, 1)$, are $[00], [01], [10], [11]$, respectively. For all $n \geq 1$, direct induction gives

$$T_f^n x = P_{x_n} \sigma^n x, \quad s_f(T_f^n x) = (0, x_{n+1} \oplus x_n), \quad (3.14)$$

where $\oplus$ is addition modulo two. In the induction step the chosen flip is $x_{n+1} \oplus x_n$; it cancels the previous flip and leaves $P_{x_{n+1}} \sigma^{n+1} x$. The first coordinate is always zero after time zero, which also proves all-time safety.

For every finite nonempty window F , with $n = |F|$, one has

$$P_{s_f}(F) = \begin{cases} 2^n, & 0 \notin F, \\ 2^{n+1}, & 0 \in F. \end{cases} \quad (3.15)$$

The four initial labels and two later labels give the upper bounds. To prove the lower bounds, prescribe any of the four initial labels if $0 \in F$, thereby fixing (x_0, x_1) ; if $0 \notin F$, set $x_0 = x_1 = 0$. At positive sampled times prescribe the desired second components d_k , and put $d_k = 0$ at the other positive times. Recursively set $x_{k+1} = x_k \oplus d_k$ for every $k \geq 1$. This constructs a full state in Q whose actual trajectory under the same selector has the required sampled controls by (3.14). This realizes every prescribed pattern on an infinite closed-loop trajectory. Equation (3.15) implies $h_A(s_f) = h^*(s_f) = \log 2$ for every A .

For the opposite bound fix any legal Borel selector s . If its first n actual controls from x are (c_k, e_k) , let $E_0 = 0$ and $E_k = e_0 \oplus \cdots \oplus e_{k-1}$. Since shifts commute with flips,

$$T_s^k x = P_{E_k} \sigma^k x.$$

Let $b(c)$ be the first bit of the word c . Safety at time k gives

$$x_k = b(c_k) \oplus E_k \quad (0 \leq k < n). \quad (3.16)$$

Thus one actual length- n control word determines at most one input prefix of length n . Every binary input prefix has an extension in Q , and the feedback is defined and safe on every such extension. Distinct prefixes must yield distinct actual control words. Hence

$$P_s(\{0, \dots, n-1\}) \geq 2^n$$

for every n and every legal s . This proves $h_{A_0}(s) \geq \log 2$ and $h^*(s) \geq \log 2$. Combining these inequalities with s_f proves (3.13), including attainment. Lemma 2.1 handles atom refinements. $\square$

The obstruction has a precise orbit interpretation. In this example $T_f(Q) = [0]$: for the reverse inclusion, given $y \in [0]$, take $x = 0y$. Therefore neither $[10]$ nor $[11]$ is visited at a positive time. The initial fibers with those labels are nonmeager, but their labels cannot be freely concatenated in the future. The map T_f is itself continuous and open, as follows by restricting to the four clopen two-symbol cylinders. It is not onto Q .

Even preservation of meager preimages does not repair this problem. For any selector s and any n , the value of $T_s^n x$ is either $\sigma^n x$ or $P_1 \sigma^n x$. Hence, for a meager set N ,

$$(T_s^n)^{-1} N \subset (\sigma^n)^{-1} N \cup (P_1 \sigma^n)^{-1} N$$

is meager. What fails is the nonempty open intersection used in Lemma 3.2 for the *selected* dynamics. Also, the same s_f gives

$$\inf_s h_A(s) \leq \log 2 = \log r - \log(3/2) \quad \text{for every } A.$$

Thus the deficit below $\log r$ persists for every sampling sequence, while $L_1 = M_1$ still holds. The exact inner minimum at other fixed sequences is not determined by this calculation.

4. INVERSE CAPACITY WHEN THE FEEDBACK CHANGES THE ORBIT

The preceding example shows why the safety cover alone is insufficient when legal controls have different successors. We now use a reference measure to bound the size of the initial-state sets carrying control names. The measure is only a tool for counting; it need not be invariant under any feedback.

The estimate concerns sets of initial states whose controls belong to prescribed lists of $k-1$ labels. If each such set has exponentially small mass, exponentially many lists are needed to cover the names of any feedback. A finite covering lemma then supplies a set of positions on which all choices of k labels occur. The mass bound is uniform over feedbacks, but these positions can depend on the rule. This is why the argument determines M_τ without yet determining L_τ .

4.1. Safe inverse-transfer operators. Fix a finite block set W of size q at period τ , with closed safety domains K_u covering Q and continuous endpoint maps $g_u : K_u \rightarrow Q$, as in (2.1). All optimizations in this section use the Borel selectors with values in this W . Let μ be a full-support Borel probability on Q and assume

$$\nu_u = (g_u)_*(\mu|_{K_u}) \ll \mu, \quad \rho_u = \frac{d\nu_u}{d\mu} \in L^\infty(\mu). \quad (4.1)$$

The function ρ_u measures how much source mass the safe branch can send to a target region. To follow this mass through several branches, define L_u by

$$\int h L_u f d\mu = \int_{K_u} h(g_u x) f(x) d\mu(x) \quad (h, f \text{ bounded Borel}). \quad (4.2)$$

Thus $L_u f$ is the density after pushing $f\mu$ through branch u and restricting the source to its safety domain. The signed pushforward is dominated in total variation by $\|f\|_\infty \nu_u$; its density exists and satisfies $|L_u f| \leq \|f\|_\infty \rho_u$. Changing f on a null set does not change the pushforward, so L_u is a positive bounded operator on $L^\infty(\mu)$ and $L_u \mathbf{1} = \rho_u$.

Fix $2 \leq k \leq q$ and set $l = k - 1$. A list box of length n is a product $S_0 \times \cdots \times S_{n-1}$ with $S_j \subset W$ and $|S_j| = l$: it allows any label in S_j at time j . For one list S , let $A_S = \sum_{u \in S} L_u$. For example, two successive lists S_0, S_1 give

$$A_{S_1} A_{S_0} \mathbf{1} = \sum_{u \in S_0, v \in S_1} L_v L_u \mathbf{1}.$$

Each summand is the terminal density obtained by following one safe two-block path. A feedback uses only part of the source set available to each path. Pushing forward the initial states whose first two labels belong to these lists therefore gives a terminal density bounded by the displayed sum. Integrating gives an upper bound on their initial mass; since μ is a probability, the L^∞ norm bounds that integral as well. We use this norm to estimate list boxes of any length:

$$b_0 = 1, \quad b_n = \max_{|S_0|=\cdots=|S_{n-1}|=l} \|A_{S_{n-1}} \cdots A_{S_0} \mathbf{1}\|_\infty. \quad (4.3)$$

The first list acts on the right. A history in (4.3) chooses one list at each time for the whole state space; it does not choose a different list at each target point. This distinction will matter in Section 4.7.

For a positive operator P , $\|P\|_{\infty \rightarrow \infty} = \|P \mathbf{1}\|_\infty$. Hence $b_{n+m} \leq b_n b_m$. In particular, one strict bound $b_h < 1$ controls all lengths: choose $b_h < \theta^h < 1$ and put

$$H = \max_{0 \leq j < h} b_j \theta^{-j}, \quad b_n \leq H \theta^n \quad (n \geq 0). \quad (4.4)$$

Indeed, write $n = ah + j$ and use submultiplicativity. The integer h is a length used to verify the estimate; the control period remains τ .

A set of words $G \subset W^n$ has a k -valued cube projection on $J \subset \{0, \dots, n-1\}$ if, for some $V \subset W$ with $|V| = k$, its projection onto J contains V^J . Thus every assignment of those k labels to the positions in J occurs in a word of G . The theorem produces such projections inside the actual feedback language, with a number of positions proportional to n .

Theorem 4.1 (Product inverse-capacity criterion). *Assume (4.1). If $b_h < 1$ for some $h \geq 1$, then there are $\alpha > 0$ and N , depending only on the branches, μ, k , such that, for every legal Borel feedback s and every $n \geq N$, its actual length- n names have a k -valued cube projection on at least αn coordinates. Consequently*

$$p_s^*(m) \geq k^m \quad (m \geq 1), \quad h^*(s) \geq \frac{\log k}{\tau}. \quad (4.5)$$

The same lower bound holds for all finite Borel atom refinements. If $k = r \geq 2$ is the minimum number of safety domains covering Q , then

$$M_\tau = \min_s h^*(s) = \frac{\log r}{\tau}, \quad (4.6)$$

both over unrefined selectors and over all finite Borel safe partitions.

4.2. From actual fibers to list covers. For a fixed feedback, group initial states by their first n controls: let

$$E_w(s) = \{x : s(T_s^j x) = w_j \ (0 \leq j < n)\}, \quad F_n(s) = \{w \in W^n : E_w(s) \neq \emptyset\}.$$

The sets $E_w(s)$ are the actual fibers. They are Borel, pairwise disjoint, and cover Q . A nonempty fiber is included even when it has measure zero. For $w = w_0 \cdots w_{n-1}$ put $L_w = L_{w_{n-1}} \cdots L_{w_0}$ and $P_F = \sum_{w \in F} L_w$.

Lemma 4.2 (Actual-fiber capacity). *For every legal Borel feedback and every $n \geq 1$,*

$$1 \leq \int P_{F_n(s)} \mathbf{1} \, d\mu. \quad (4.7)$$

If $F_n(s)$ is covered by t boxes $S_0^a \times \cdots \times S_{n-1}^a$ whose lists have size at most l , then $1 \leq tb_n$.

Proof. Follow the mass in one fiber through its prescribed branches. Safety means that none is lost at a restriction to K_{w_j} . Dropping the fiber restriction at the start can only increase the resulting density.

Fix a nonempty fiber E_w and let η_j be the pushforward of $\mu|_{E_w}$ after its first j actual blocks. Set $f_0 = \mathbf{1}_{E_w}$. We claim inductively that $\eta_j = f_j \mu$ and $f_{j+1} = L_{w_j} f_j$. Indeed, η_j is concentrated on K_{w_j} , and for every bounded Borel h ,

$$\int h \, d\eta_{j+1} = \int_{K_{w_j}} h(g_{w_j} x) f_j(x) \, d\mu(x) = \int h L_{w_j} f_j \, d\mu.$$

The boundedness of each L_u permits the induction. None of these restrictions removes mass, because all states in E_w follow the specified safe word. Thus $\int L_w \mathbf{1}_{E_w} \, d\mu = \mu(E_w)$. Positivity gives $L_w \mathbf{1}_{E_w} \leq L_w \mathbf{1}$; summing over the disjoint fibers proves (4.7). The set $F_n(s)$ includes null but nonempty fibers as well.

Complete every nonempty list to size l . Distributivity and positivity give

$$P_{F_n(s)} \mathbf{1} \leq \sum_{a=1}^t A_{S_{n-1}^a} \cdots A_{S_0^a} \mathbf{1}.$$

Repeated boxes and overlaps only enlarge this upper bound. Unsafe words in a box are subject to the original safety gates in L_u . Integrating and using (4.3) yields $1 \leq tb_n$. $\square$

4.3. The finite combinatorial input and its application. The preceding mass estimate gives a lower bound on the number of list boxes needed to cover the names. The finite covering lemma we will use is stated for a different kind of box: at each position it omits one letter from a fixed k -element alphabet. We compare these two covering numbers before applying the lemma.

For $F \subset W^n$, let $C_l(F)$ be its minimum covering number by l -list boxes. For a k -element set $V \subset W$, let $C_V(F)$ be the minimum number of boxes of the form

$$\prod_{j=0}^{n-1} (W \setminus \{v_j\}), \quad v_j \in V,$$

needed to cover F . This is a cover, since $k \geq 2$.

We use the following finite covering lemma. For every $k \geq 2$ and $b > 0$ there are $c > 0$ and N such that if $G \subset \{0, 1, \dots, k\}^n$, $n \geq N$, needs at least e^{bn} boxes omitting one nonzero letter per coordinate, then some $J \subset \{0, \dots, n-1\}$ with $|J| \geq cn$ has

$$G|_J \supset \{1, \dots, k\}^J. \quad (4.8)$$

This is Kerr–Li’s finite covering lemma [8, Lemma 3.3]. Huang and Ye recover the same conclusion from their sharper estimates in [4, Lemma 4.1 and Remark 4.2, pp. 1701–1705]. Its constants depend only on b, k ; its use here requires no dynamical assumptions on the closed loop.

To pass from C_l to one of the numbers C_V , choose a minimum omitted-letter cover for every V and intersect all these covers. The intersections cover F , with at most $\prod_V C_V(F)$ members. In a nonempty intersection each coordinate contains at most $k-1$ letters: if it contained a k -set V_0 , its factor from the V_0 cover would have omitted a letter of V_0 . Hence

$$C_l(F) \leq \prod_{V \in \binom{W}{k}} C_V(F). \quad (4.9)$$

For actual names Lemma 4.2 gives $C_l(F_n(s)) \geq b_n^{-1}$. Under (4.4), for all sufficiently large n this lower bound is at least $e^{\eta n/2}$, where $\eta = -\log \theta > 0$. With $J_0 = \binom{q}{k}$, (4.9) implies that some V satisfies

$$C_V(F_n(s)) \geq e^{\eta n/(2J_0)}.$$

Code the members of V by $1, \dots, k$ and all other letters by 0. Omitted-letter boxes pull back exactly to the boxes defining C_V , so the coded set has precisely that covering number. Applying (4.8) and undoing this coding yields $F_n(s)|_J \supset V^J$. No distinct members of V were merged.

Proof of Theorem 4.1. The preceding argument supplies α, N independent of s . Given m , choose n so large that $\alpha n \geq m$ and retain m of the independent positions. All k^m patterns are actual feedback names, proving (4.5) with the prescribed normalization by $m\tau$.

For $k = r$, choose a minimum subcover $K_{u_1}, \dots, K_{u_r}$ and select u_j on $K_{u_j} \setminus \bigcup_{i < j} K_{u_i}$. This is a Borel state function, legal at every point. Each chosen block stays safe at its intermediate times and ends in Q ; applying the same rule at the next state gives infinite safety. It uses r labels and has at most r^m patterns on any m positions. This proves (4.6) and attainment. For a refined partition, projection to its physical control labels is onto the actual coarse patterns and preserves the dynamics. The lower bound therefore passes to every refinement, while the unrefined selector remains an available upper-bound witness in the enlarged class. $\square$

The conclusion controls M_τ . To control L_τ by this method, one would also need to choose the independent positions in common for all feedbacks. The covering lemma provides no such choice: its J and V may depend on both s and n .

4.4. Single-step scalar and joint-density criteria. Define

$$\Theta_k = \left\| \max_{|S|=k-1} \sum_{u \in S} \rho_u \right\|_\infty. \quad (4.10)$$

If $\Theta_k < 1$, then $b_1 = \Theta_k < 1$ and Theorem 4.1 applies. If the branch capacities obey $\mu(K_u \cap g_u^{-1}E) \leq c_u \mu(E)$ for every Borel E , then $\rho_u \leq c_u$ almost everywhere, and the sum of the largest $k-1$ of the c_u bounds Θ_k . Thus a scalar-capacity sum smaller than one is a sufficient special case. The order of taking the branch supremum and summing matters: joint densities may pass the test while the individual optimal scalar bounds do not.

Appendix B gives a direct quantitative estimate in this single-step case. The strict inequality is essential: Proposition B.2 gives systems with $r = 3$, $\Theta_3 \geq 1$ arbitrarily close to one, and $M_1 = \log 2$.

4.5. The excluded-letter system with different successors. Return to the excluded-letter cover, but add the chosen control to every coordinate of the shifted state. The safety cover is unchanged; the trajectory now depends on the feedback. This gives a simple application of the capacity theorem and shows what information it leaves open.

Let $q \geq 3$, $G = \mathbb{Z}/q\mathbb{Z}$, and $Q = G^{\mathbb{N}_0}$. Write P_u for coordinatewise addition of u . On $X = Q \sqcup \{\dagger\}$ use the continuous controls

$$f_u(x) = P_u \sigma x \quad \text{if } x_0 \neq u, \quad f_u(x) = \dagger \quad \text{otherwise,} \quad f_u(\dagger) = \dagger.$$

Their safety domains are the excluded-letter cylinders $K_u = \{x : x_0 \neq u\}$, so the minimum cover cardinality is two.

Corollary 4.3. *For this system, at period one and over all legal Borel feedbacks and finite atom refinements,*

$$M_1 = \min_s h^*(s) = \log 2, \quad \log \frac{q}{q-1} \leq L_1 \leq \log 2, \quad h_{\text{inv}}(Q) = \log \frac{q}{q-1}.$$

There is a clopen feedback with $P_s(F) = 2^{|F|}$ for every finite window F .

Proof. Use the uniform Bernoulli probability on Q . The safe inverse operator is

$$(L_u f)(y) = \frac{1}{q} \sum_{a \neq u} f(a P_{-u} y), \quad L_u \mathbf{1} = \frac{q-1}{q} \mathbf{1}.$$

Products of any single-label lists therefore have norm $((q-1)/q)^n$. Theorem 4.1, with $k = r = 2$, gives $M_1 = \log 2$.

For a control word $w = (w_0, \dots, w_{n-1})$ produced by s , put $E_k = \sum_{j < k} w_j$ and $v_k = w_k - E_k$, with arithmetic in G . The actual orbit satisfies $T_s^k x = P_{E_k} \sigma^k x$, so its initial fiber lies in

$$K(w) = \{x : x_k \neq v_k \ (0 \leq k < n)\}.$$

This is also the complete open-loop safety domain of w . Each such domain contains exactly $(q-1)^n$ input prefixes. Every input prefix extends to a state with an infinite safe feedback trajectory; hence the actual length- n word count is at least $(q/(q-1))^n$. Consecutive sampling gives the stated lower bound for L_1 .

Define $s_0(x) = 1$ if $x_0 = 0$, and $s_0(x) = 0$ otherwise. This is clopen and safe on all of Q . Given any infinite binary sequence d , let $E_k = \sum_{j < k} d_j$ and let $b_k = 0$ when $d_k = 1$, $b_k = 1$ when $d_k = 0$. The state with coordinates $x_k = b_k - E_k$ has current first coordinate b_k at time k , by induction. Its feedback name is therefore d . This proves the finite-window equality and the upper bound for L_1 with the same feedback for every sampling sequence.

The map $w \mapsto v$ is triangular and bijective: recover $w_k = v_k + E_k$, $E_{k+1} = 2E_k + v_k$, starting with $E_0 = 0$. This requires no inverse of 2, and includes even q . Thus the open-loop spanning number is the least number $c_q(n)$ of boxes $\prod_{k < n} (G \setminus \{v_k\})$ covering G^n . Writing $p = ((q-1)/q)^n$, the counting lower bound and the elementary random-cover upper bound give

$$p^{-1} \leq c_q(n) \leq \lceil p^{-1}(n \log q + 1) \rceil.$$

Indeed, $t = \lceil p^{-1}(n \log q + 1) \rceil$ independent uniform boxes leave some input uncovered with probability at most $q^n(1-p)^t \leq q^n e^{-pt} \leq e^{-1} < 1$. A covering choice exists, and removal of repetitions can only decrease its size. Taking normalized logarithms proves the invariance-entropy formula. Lemma 2.1 supplies the refinement statement. $\square$

The safe spanning number is a lower bound for the actual feedback word count, not an upper bound. For example, when $q = 3$, the three safe words 00, 12, 21 cover all two-symbol inputs: their excluded vectors are 00, 11, 22. The clopen rule $s(x) = \min(G \setminus \{x_0, x_1\})$ repeatedly selects the first control of a covering word and is safe everywhere. Starting from $x = (1, 1, 0, 0, \dots)$, however, it selects 0 and then 2, giving the word 02 outside that cover.

The subsequent state is 2^∞ and all subsequent controls are 0, so the witness has an infinite safe continuation. Its prefix fibers obey

$$H_s(uw) = s^{-1}(u) \cap (P_u\sigma)^{-1}H_s(w), \quad H_s(w) \subset K(w).$$

These compatibility equations are absent from a separately chosen spanning family. The lower and upper bounds for L_1 remain distinct; neither their equality nor a strict minimax gap is asserted for this cyclic system.

4.6. A smooth joint-density example. The joint-density test can be sharper than taking a separate supremum for each branch: peaks at different target points need not add. On $Q = [0, 1]$ take three full inverse branches with

$$\begin{aligned} l_u &= \frac{19u}{60}, & \rho_u(y) &= \frac{11}{30} + \frac{7}{30} \cos(2\pi(y - u/3)), \\ \psi_u(y) &= l_u + \frac{11}{30}y + \frac{7}{60\pi} [\sin(2\pi(y - u/3)) - \sin(-2\pi u/3)]. \end{aligned} \quad (4.11)$$

Here $g_u = \psi_u^{-1}$, for $u = 0, 1, 2$. Their derivatives lie between $2/15$ and $3/5$, so the branches are strictly increasing and expanding in the forward direction. The closed safety intervals are $[0, 22/60]$, $[19/60, 41/60]$, and $[38/60, 1]$. They cover Q and each is needed: the middle interval alone covers the open gap between $22/60$ and $38/60$, while the two endpoints require the first and third intervals. Hence $r = 3$.

With Lebesgue probability, the densities are exactly $\psi'_u = \rho_u$. The three cosine terms sum to zero. The sum of two is the negative of the remaining cosine, whose maximum is one; therefore

$$\Theta_3 = \frac{22}{30} + \frac{7}{30} = \frac{29}{30} < 1.$$

Each branch has optimal scalar bound $3/5$, since its continuous density reaches that value. The largest two scalar bounds sum to $6/5$. The joint criterion consequently gives $M_1 = \log 3$ although that Lebesgue scalar criterion fails.

To place these interval branches in a continuous compact ambient system, take $X = [-1, 2]$. Extend each g_u to the left of its safety interval $[a_u, b_u]$ by $-\min\{a_u - x, 1\}$ and to its right by $1 + \min\{x - b_u, 1\}$. At the endpoints these formulas match 0 and 1. Outside the closed safety interval the image lies outside Q , so this extension changes neither safety domains nor legal dynamics. The same construction applies to full monotone interval branches elsewhere in the paper.

This example compares the two single-step tests for Lebesgue probability. The next example addresses a different question: can a product succeed when no choice of full-support reference probability makes the single-step test work?

4.7. Product contraction without a single-step reference test. Single-step contraction can fail for every reference probability even when a short product contracts. In the following example, densities depend on one coordinate, so the product norm can be computed exactly.

Let $Q = (\mathbb{Z}/3\mathbb{Z})^{\mathbb{N}_0}$, let P_u add u to each coordinate, and use $g_u = P_u\sigma$ at period one. Put

$$K_0 = [02] \cup [12] \cup [22], \quad K_1 = [00] \cup [20], \quad K_2 = [01] \cup [10] \cup [11] \cup [12] \cup [21].$$

The only overlap is $[12]$. The cylinders $[02]$, $[00]$, $[01]$ require controls 0, 1, 2, respectively, so $r = 3$. On $[12]$, choosing 0 or 2 changes the successor's first letter from 2 to 1. Every Borel mixture on this entire overlap is allowed. Illegal controls go to an isolated fixed cemetery point; the resulting maps on $Q \sqcup \{\dagger\}$ are continuous.

For fair Bernoulli probability, the operators leave first-coordinate functions invariant and have matrices $M_u/3$, where

$$M_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \quad M_1 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

This follows from splitting the source measure by its first letter, and retaining exactly the allowed two-letter cylinders. For a positive product the full L^∞ norm equals its norm on $\mathbf{1}$, whose entire orbit stays in this invariant space. Thus this finite representation computes the full product norms, not an approximation to arbitrary Borel functions.

Let $A = M_0 + M_1$, $B = M_0 + M_2$, $C = M_1 + M_2$. The vectors after two of these unnormalized factors are

$$\begin{pmatrix} 0, 3, 5 \\ 5, 2, 7 \\ 5, 5, 2 \end{pmatrix} \quad \begin{pmatrix} 0, 7, 8 \\ 8, 1, 9 \\ 8, 8, 1 \end{pmatrix} \quad \begin{pmatrix} 0, 4, 7 \\ 7, 3, 10 \\ 7, 7, 3 \end{pmatrix}.$$

They are coordinatewise dominated by one of the vectors in

$$V = \{(0, 7, 8), (7, 3, 10), (7, 7, 3), (8, 1, 9), (8, 8, 1)\}.$$

For $v = (p, q', r')$, $Av = (0, p + r', p + q' + r')$, $Bv = (p + q' + r', q', p + 2q' + r')$, and $Cv = (p + q' + r', p + q' + r', q')$. For the five vectors of V , their largest resulting coordinate is, respectively, 22, 23, 24, 19, 25. Also $BCB\mathbf{1} = (17, 8, 25)$. Consequently $b_3 = 25/27 < 1$. Submultiplicativity, rather than an extrapolation from finite calculations, controls all lengths. Theorem 4.1 gives $M_1 = \log 3$, and every unrefined feedback has that maximal-pattern rate. An explicit safe selector chooses 0 if $x_1 = 2$, otherwise 1 if $x_1 = 0$ and $x_0 \neq 1$, and otherwise 2.

On the three target first-letter cylinders the densities are $(0, 0, 1)$, $(0, 2/3, 1/3)$, and $(1, 0, 1/3)$. Hence $\Theta_3 = 4/3$. This failure persists under *every* full-support probability ν . If one of the safe branch measures has no bounded ν -density, the single-step criterion already fails its prerequisites. Otherwise $g_0(K_0) \subset [2]$ and $g_1(K_1) \subset [1]$ give disjoint density supports. At most two densities are nonzero at a point, so their largest two sum to their total. Its integral is

$$\sum_{u=0}^2 \nu(K_u) = 1 + \nu([12]) > 1.$$

Thus $\Theta_3(\nu) > 1$ for every such ν , including singular or atomic probabilities. This excludes both original single-step tests, but not all possible changes of presentation or iteration.

To compare global lists with state-dependent maximization, put

$$\Phi f = \max_{|S|=2} \sum_{u \in S} L_u f \quad (f \geq 0).$$

On nonnegative functions the disjoint supports give $\Phi = \sum_u L_u$. Under fair measure its first-letter matrix is $\frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix}$, with positive eigenvector $(1, 1, (\sqrt{13} - 1)/2)$ and eigenvalue $(3 + \sqrt{13})/6 > 1$. Hence $\|\Phi^n \mathbf{1}\|_\infty$ grows, while every history of *global* two-label lists decays. Maximization by state and composition cannot be interchanged.

5. COUNTABLE LANGUAGES AND COMMON SAMPLING

The capacity theorem chooses good observation positions separately for each feedback. If only countably many name languages occur, they can be served in turn by one increasing sequence. The main point is to move on to later coordinates without translating a name language: translation can lose patterns for a one-sided system. We instead delete the old coordinates from a much larger maximizing window.

5.1. Simultaneous attainment for pattern systems. A *finite-alphabet pattern system* is a nonempty set $\Omega \subset E^{\mathbb{N}_0}$ with E finite and nonempty. No topology or shift invariance is imposed. For finite $F \subset \mathbb{N}_0$, write

$$P_\Omega(F) = \#\{x|_F : x \in \Omega\}, \quad p_\Omega^*(n) = \max_{\substack{F \subset \mathbb{N}_0 \\ \#F=n}} P_\Omega(F).$$

With a fixed scale $\tau > 0$, define

$$h^*(\Omega; \tau) = \lim_{n \rightarrow \infty} \frac{1}{n\tau} \log p_\Omega^*(n),$$

$$h_{\text{seq}}^A(\Omega; \tau) = \limsup_{n \rightarrow \infty} \frac{1}{n\tau} \log P_\Omega(\{a_1, \dots, a_n\}).$$

The maximum exists: its possible values form a nonempty subset of the finite integer set $\{1, \dots, (\#E)^n\}$.

We use $P_\Omega(\emptyset) = 1$ when needed. Taking the product-topology closure of Ω changes none of its finite projections: a cylinder meeting $\overline{\Omega}$ already meets Ω . Thus dispensing with closedness is only a matter of formulation. The absence of shift invariance, in contrast, matters to the placement of the coordinate sets.

Lemma 5.1 (Pattern submultiplicativity and deletion). *Let $\Omega \subset E^{\mathbb{N}_0}$ be a pattern system and $q = \#E$. For $m, n \in \mathbb{N}$ and finite $B \subset F \subset \mathbb{N}_0$,*

$$p_\Omega^*(m+n) \leq p_\Omega^*(m)p_\Omega^*(n), \quad (5.1)$$

$$P_\Omega(F) \leq q^{\#(F \setminus B)} P_\Omega(B). \quad (5.2)$$

If $B \subset G$, then $P_\Omega(B) \leq P_\Omega(G)$. Consequently,

$$h^*(\Omega; \tau) = \inf_{n \geq 1} \frac{1}{n\tau} \log p_\Omega^*(n), \quad \log p_\Omega^*(n) \geq n\tau h^*(\Omega; \tau). \quad (5.3)$$

Proof. Split any $(m+n)$ -element set into disjoint subsets of sizes m and n . Restriction sends a pattern on the union injectively to the pair of its two restrictions, proving (5.1). Restriction from F to B has fibers of cardinality at most $q^{\#(F \setminus B)}$, which proves (5.2). If $B \subset G$, every pattern on B extends to a pattern on G using the same element of Ω , giving monotonicity. Fekete's lemma applied to $\log p_\Omega^*(n)$ proves (5.3). $\square$

For comparison, let $\Omega = \{0^\infty, 10^\infty\}$. This set is a closed one-sided subshift, but $P_\Omega(\{0\}) = 2$ and $P_\Omega(\{m\}) = 1$ for $m \geq 1$. One cannot simply translate a maximizing set. Deleting a finite initial interval has a different advantage: the number of lost coordinates is bounded independently of the size of the maximizing set. Inequality (5.2) turns this bound into a controlled information loss.

Theorem 5.2 (Simultaneous attainment). *Fix $\tau > 0$. For every countable family $(\Omega_j)_{j \geq 1}$ of finite-alphabet pattern systems, there is one $A \in \mathcal{S}$ such that*

$$h_{\text{seq}}^A(\Omega_j; \tau) = h^*(\Omega_j; \tau) \quad (j \geq 1). \quad (5.4)$$

The sets Ω_j may be arbitrary nonempty subsets of their finite-alphabet sequence spaces; shift invariance is not required.

Proof. At each stage we choose a large maximizing window for one language and delete all positions preceding the end of the previous stage. There are two losses to control: the deleted coordinates, and the earlier observations still present in the denominator. Making the new window large controls both. Each language receives infinitely many stages, which suffices for its limsup.

Let E_j be the alphabet of Ω_j , put $q_j = \#E_j$ and $h_j^* = h^*(\Omega_j; \tau)$. For every A and n ,

$$P_{\Omega_j}(\{a_1, \dots, a_n\}) \leq p_{\Omega_j}^*(n),$$

so (5.3) gives $h_{\text{seq}}^A(\Omega_j; \tau) \leq h_j^*$.

Choose an enumeration of stages

$$(j_k)_{k \geq 1} = (1; 1, 2; 1, 2, 3; 1, 2, 3, 4; \dots),$$

where semicolons only separate finite groups. Every positive integer occurs infinitely often. Choose $\eta_k > 0$ with $\eta_k \rightarrow 0$. We recursively construct nonempty finite sets B_k , each lying strictly to the right of all previous ones. Put

$$G_{k-1} = \bigcup_{i < k} B_i, \quad r_{k-1} = \#G_{k-1}, \quad M_{k-1} = \max G_{k-1},$$

with $r_0 = 0$ and $M_0 = -1$.

If $h_{j_k}^* > 0$, choose n_k so large that

$$n_k > M_{k-1} + 1, \quad \frac{(M_{k-1} + 1) \log q_{j_k}}{n_k \tau} < \eta_k, \quad \frac{r_{k-1}}{n_k} < \eta_k. \quad (5.5)$$

Both M_{k-1} and r_{k-1} are fixed at this stage. The second inequality pays for deleting up to $M_{k-1} + 1$ positions; the third makes the earlier r_{k-1} observations negligible. Choose an n_k -element F_k attaining $p_{\Omega_{j_k}}^*(n_k)$ and set

$$B_k = F_k \cap \{M_{k-1} + 1, M_{k-1} + 2, \dots\}.$$

At most $M_{k-1} + 1$ coordinates are deleted, and the first condition in (5.5) makes B_k nonempty. If $h_{j_k}^* = 0$, set $B_k = \{M_{k-1} + 1\}$.

Let A be the increasing enumeration of $\bigcup_{k \geq 1} B_k$. Fix j with $h_j^* > 0$ and restrict to the infinitely many stages with $j_k = j$. Write $b_k = \#B_k$ and $R_k = r_{k-1} + b_k = \#G_k$; then G_k consists of the first R_k entries of A . By deletion, (5.3), and monotonicity,

$$\begin{aligned} \log P_{\Omega_j}(G_k) &\geq \log P_{\Omega_j}(B_k) \\ &\geq \log p_{\Omega_j}^*(n_k) - (M_{k-1} + 1) \log q_j \\ &\geq n_k \tau h_j^* - (M_{k-1} + 1) \log q_j. \end{aligned} \quad (5.6)$$

Also $R_k \leq r_{k-1} + n_k$. At all sufficiently late such stages the last line of (5.6) is positive. Dividing by $R_k \tau$ and using (5.5) yields the explicit endpoint estimate

$$\frac{1}{R_k \tau} \log P_{\Omega_j}(G_k) > \frac{h_j^* - \eta_k}{1 + \eta_k}. \quad (5.7)$$

For a fixed j , the indices k with $j_k = j$ tend to infinity, hence $\eta_k \rightarrow 0$ along those stages. Since every B_i is nonempty, $R_k \geq k \rightarrow \infty$. The limsup along these endpoints is at least h_j^* , proving the lower bound. If $h_j^* = 0$, the universal upper bound and nonnegativity give equality for every A . This proves (5.4) for all j . $\square$

Remark 5.3 (What is attained). The theorem asserts equality of limsup rates. It does not assert a common subsequence of endpoints on which every system is simultaneously close to its maximal rate. The good endpoints used for different j can be different. Nor does it assert convergence of the normalized counts. For example, take $\tau = 1$, let $B \subset \mathbb{N}_0$ and its complement both be infinite, and let Ω_B consist of all binary sequences supported on B . For any schedule, the two normalized logarithmic counts for Ω_B and $\Omega_{\mathbb{N}_0 \setminus B}$ add to $\log 2$ at every n , although both maximal-pattern rates are $\log 2$. They cannot both converge to $\log 2$. The two limsups, however, can both equal $\log 2$.

Exact finite-level maximization is also a stronger and generally false requirement. Put

$$\Omega = \{0x : x \in \{0, 1\}^{\mathbb{N}_0}\} \cup \{10^\infty, 20^\infty\} \subset \{0, 1, 2\}^{\mathbb{N}_0}.$$

The unique maximizing singleton is $\{0\}$, with three patterns. For a three-element set containing zero there are six patterns, whereas any three positive coordinates realize eight patterns. No increasing sequence maximizes the counts at both levels $n = 1$ and $n = 3$.

5.2. Partition names. To apply the preceding theorem to invariant partitions, use the atom itineraries introduced in Section 2. We now allow an arbitrary control set U ; each partition still assigns only finitely many blocks. Write a finite invariant partition as $\mathcal{C} = (\{D_i : i \in I_{\mathcal{C}}\}, \tau, \nu)$, with nonempty Borel atoms D_i and assigned blocks $\nu(D_i) \in U^\tau$ satisfying

$$\varphi(j, x, \nu(D_i)) \in Q \quad (x \in D_i, 0 \leq j \leq \tau). \quad (5.8)$$

Let $\mathfrak{C}_{\text{Bor}}(Q, \tau)$ denote this class and $\mathfrak{C}_{\text{clop}}(Q, \tau)$ its clopen subclass. For variable periods we use

$$\mathfrak{C}_{\text{Bor}}(Q) = \bigcup_{\tau \geq 1} \mathfrak{C}_{\text{Bor}}(Q, \tau), \quad \inf \emptyset = +\infty.$$

The state map is $T_{\mathcal{C}}x = \varphi(\tau, x, \nu(D_i))$ on D_i . Let $\iota_{\mathcal{C}}(x)$ be its atom itinerary and $W_N(\mathcal{C}; Q)$ the set of length- N prefixes. Thus $a_0 \cdots a_{N-1}$ belongs to $W_N(\mathcal{C}; Q)$ exactly when some $x \in Q$ satisfies

$$\varphi(j\tau, x, \nu(D_{a_0}) \cdots \nu(D_{a_{N-1}})) \in D_{a_j} \quad (0 \leq j < N). \quad (5.9)$$

These are Kawan's admissible words [7, Definition 2.9]; the induced map is also used in [11, Section 2]. Condition (5.8) ensures safety between successive block boundaries. After the last block the same rule supplies the next one, so every finite witness continues indefinitely. A suffix is witnessed by the state at its starting block. In particular,

$$|W_{M+N}(\mathcal{C}; Q)| \leq |W_M(\mathcal{C}; Q)| |W_N(\mathcal{C}; Q)|. \quad (5.10)$$

We shall sometimes use the compact name space

$$\Sigma_{\mathcal{C}} = \{a \in I_{\mathcal{C}}^{\mathbb{N}_0} : a|_{[0, N-1]} \in W_N(\mathcal{C}; Q) \text{ for every } N\}. \quad (5.11)$$

It is closed, and suffix closure gives $\sigma\Sigma_{\mathcal{C}} \subset \Sigma_{\mathcal{C}}$. The cylinder description yields

$$\Sigma_{\mathcal{C}} = \overline{\iota_{\mathcal{C}}(Q)}. \quad (5.12)$$

Every cylinder meeting this closure already contains an itinerary. Hence its finite patterns are exactly the actual patterns, and our earlier notation becomes

$$P_{\mathcal{C}}(F) = |\{a|_F : a \in \Sigma_{\mathcal{C}}\}|, \quad (5.13)$$

$$p_{\mathcal{C}}^*(n) = \max_{|F|=n} P_{\mathcal{C}}(F), \quad (5.14)$$

$$h^*(\mathcal{C}; Q) = h^*(\Sigma_{\mathcal{C}}; \tau), \quad (5.15)$$

$$h_{\text{seq}}^A(\mathcal{C}; Q) = h_{\text{seq}}^A(\Sigma_{\mathcal{C}}; \tau). \quad (5.16)$$

We also need the consecutive-name rate

$$h(\mathcal{C}; Q) = \lim_N \frac{\log |W_N(\mathcal{C}; Q)|}{N\tau}, \quad (5.17)$$

whose limit follows from (5.10).

Remark 5.4 (Infinite names in the closure). For clopen atoms the maps $T_{\mathcal{C}}$ and $\iota_{\mathcal{C}}$ are continuous, so compactness gives $\Sigma_{\mathcal{C}} = \iota_{\mathcal{C}}(Q)$. With Borel atoms, some infinite names in the closure can lack a single state witness. For example, take $Q = [0, 1]$, one control $f(x) = \min\{2x, 1\}$, and atoms $D_0 = (0, 1/2)$, $D_1 = Q \setminus D_0$. The state 2^{-N-1} realizes 0^N , but 0^∞ has no witness: zero is not in D_0 , and each positive point eventually leaves it. The finite counts in (5.13) are nevertheless actual trajectory counts, by the cylinder argument above.

Corollary 5.5 (Fixed-language attainment). *For every finite-alphabet pattern system Ω and every $\tau > 0$,*

$$h^*(\Omega; \tau) = \max_{A \in \mathcal{J}} h_{\text{seq}}^A(\Omega; \tau). \quad (5.18)$$

The same identity holds for every finite Borel invariant partition $\mathcal{C}$ after putting $\Omega = \Sigma_{\mathcal{C}}$.

5.3. Feedback classes with countably many languages. Fix a compact controlled-invariant set Q and one period $\tau \in \mathbb{N}$. For two invariant partitions $\mathcal{C}$ and $\mathcal{D}$ of this period, write $\mathcal{C} \sim_{\text{lang}} \mathcal{D}$ if there is a bijection $\pi : I_{\mathcal{C}} \rightarrow I_{\mathcal{D}}$ such that, for every $N \in \mathbb{N}$,

$$\pi^N(W_N(\mathcal{C}; Q)) = W_N(\mathcal{D}; Q), \quad (5.19)$$

where π^N acts coordinatewise. This is *language equivalence*. Identity, inverse, and composition of the alphabet bijections show that $\sim_{\text{lang}}$ is an equivalence relation. If $F \subset \mathbb{N}_0$ is nonempty and finite and $m = \max F$, then (5.19) at length $m+1$, followed by projection onto F , gives a bijection between the two F -pattern sets. Thus language-equivalent partitions have identical finite-coordinate pattern counts and identical sequence and maximal-pattern complexities.

Fix a nonempty class $\mathfrak{C} \subset \mathfrak{C}_{\text{Bor}}(Q, \tau)$. Every individual rate is nonnegative and finite, so its infimum over this class is finite even without a uniform bound on the numbers of atoms.

The simultaneous-attainment theorem applies once there are only countably many such languages, up to renaming their letters. Taking infima along its common sequence gives the reverse of the elementary minimax inequality. This yields the following direct corollary.

Corollary 5.6 (Minimax over countably many language types). *Let $\emptyset \neq \mathfrak{C} \subset \mathfrak{C}_{\text{Bor}}(Q, \tau)$ and suppose that the quotient $\mathfrak{C}/\sim_{\text{lang}}$ is finite or countable. There is one $A \in \mathcal{S}$ such that*

$$h_{\text{seq}}^A(\mathcal{C}; Q) = h^*(\mathcal{C}; Q) \quad (\mathcal{C} \in \mathfrak{C}). \quad (5.20)$$

Consequently,

$$\max_{B \in \mathcal{S}} \inf_{\mathcal{C} \in \mathfrak{C}} h_{\text{seq}}^B(\mathcal{C}; Q) = \inf_{\mathcal{C} \in \mathfrak{C}} h^*(\mathcal{C}; Q). \quad (5.21)$$

The outer maximum is attained. The pattern infimum and the inner infimum at the common attaining schedule are minima, by Corollary A.2.

Proof. Choose one representative $\mathcal{C}_j$ of each language-equivalence class. Enumerate the representatives, repeating them periodically when the quotient is finite, and apply Theorem 5.2 to the resulting countable family $\Sigma_{\mathcal{C}_j}$, using the common scale τ . The resulting A gives equality for each representative. Language equivalence preserves all finite pattern counts, so it gives (5.20) for every $\mathcal{C} \in \mathfrak{C}$. Taking the infimum over $\mathcal{C}$ gives equality at this A .

For any $B \in \mathcal{S}$, Corollary 5.5 gives $h_{\text{seq}}^B(\mathcal{C}; Q) \leq h^*(\mathcal{C}; Q)$ for every $\mathcal{C}$. Taking first the infimum over $\mathcal{C}$ and then the supremum over B proves the reverse minimax inequality and hence (5.21). If the pattern infimum is zero, the pointwise upper bound and nonnegativity in fact give $\inf_{\mathcal{C} \in \mathfrak{C}} h_{\text{seq}}^B(\mathcal{C}; Q) = 0$ for every $B \in \mathcal{S}$, without any countability assumption. $\square$

Language equivalence allows alphabet renumbering while preserving all finite pattern counts. It does not merge distinct atoms merely because their assigned controls agree. The countability assumption concerns the languages of the chosen partition class at the fixed period.

Proposition 5.7 (Countability of clopen feedbacks). *Assume that Q is compact metrizable and that the control alphabet U is countable. Then $\mathfrak{C}_{\text{clon}}(Q, \tau)$ is countable. Hence, if this class is nonempty, Corollary 5.6 applies to it. In particular this holds when U is a finite digital alphabet.*

Proof. A compact metrizable space has a countable base. Every clopen subset is compact and hence is a finite union of basic open sets contained in it. Thus Q has only countably many clopen subsets and only countably many finite clopen partitions. Since τ is finite, U^τ is countable. A finite partition admits only countably many block assignments. The invariant assignments form a subcollection. $\square$

Corollary 5.6 fixes τ . It does not produce one common physical-time schedule for partitions of different periods and proves no minimax identity over $\mathfrak{C}_{\text{Bor}}(Q)$.

The countability hypothesis has concrete limitations. Appendix F develops the elementary obstruction of Example F.1 into autonomous constructions. In particular, Proposition F.3 uses four distinct controls and has $L_1 = M_1 = \log 2$, although no sequence attains every

feedback's own maximal-pattern rate. It therefore separates pointwise common attainment from a common full-class minimum.

6. COMPARISON WITH CLASSICAL AND INVARIANCE ENTROPY

A fixed-period pattern rate can be positive when invariance entropy vanishes. We first place the counts beside the usual entropy notions and explain what happens when the control period varies. A symbolic realization then gives an explicit separation at period one.

6.1. Classical entropy and sequence entropy. For a continuous map $T : Z \rightarrow Z$ of a compact metric space and a finite open cover $\mathcal{V}$, replace the name count by

$$c_{\mathcal{V}}(F) = N \left(\bigvee_{j \in F} T^{-j} \mathcal{V} \right),$$

where N denotes minimum subcover cardinality. Consecutive windows, prefixes of A , and maximizing n -element windows give, respectively,

$$\begin{aligned} h_{\text{top}}(T, \mathcal{V}) &= \lim_n n^{-1} \log c_{\mathcal{V}}(\{0, \dots, n-1\}), \\ h_{\text{top}}^A(T, \mathcal{V}) &= \limsup_n n^{-1} \log c_{\mathcal{V}}(\{a_1, \dots, a_n\}), \\ h_{\text{top}}^*(T, \mathcal{V}) &= \lim_n n^{-1} \log \max_{|F|=n} c_{\mathcal{V}}(F). \end{aligned}$$

Submultiplicativity gives the first and third limits. Taking the supremum of the first over finite open covers defines $h_{\text{top}}(T)$.

For homeomorphisms, Huang and Ye prove

$$h_{\text{top}}^*(T, \mathcal{V}) = \max_A h_{\text{top}}^A(T, \mathcal{V}) \quad (6.1)$$

[4, Theorem 2.1(1)]. Their proof places successive maximizing windows farther to the right by translation. Surjectivity preserves cover counts under that translation. The deletion argument in Section 5 instead applies to arbitrary name sets, including those of non-surjective maps.

For a clopen partition, $c_{\mathcal{V}}(F)$ is the number of nonempty atoms of the join, so it is precisely a name count. For an overlapping cover it is a minimum covering number, as illustrated by Corollary 3.4. A further distinction is needed for measurable entropy: Shannon entropy weights the atoms by an invariant probability, whereas our counts retain every nonempty atom. Example E.2 shows that even a countable set of initial states can change the counting rate.

On a subshift the coordinate partition gives the name counts used here. For ordinary entropy its consecutive joins refine every fixed cylinder partition without changing the exponential rate. Sparse observations behave differently: observing a window of several coordinates at each selected time can increase the rate per observation. Accordingly, we write “coord” for the coordinate rates below; we do not take a supremum over open covers.

6.2. Comparison with Kawan's invariance entropy. For every invariant partition, consecutive coordinates are one permissible choice in the maximal-pattern count. Therefore

$$h(\mathcal{C}; Q) \leq h^*(\mathcal{C}; Q) \leq \frac{\log \#I_{\mathcal{C}}}{\tau}. \quad (6.2)$$

Kawan's Theorem 2.3, with natural logarithms, gives

$$h_{\text{inv}}(Q) = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Q)} h(\mathcal{C}; Q). \quad (6.3)$$

Its hypotheses are satisfied by the full-control discrete-time systems used here: X is metrizable, each controlled map is continuous, and Q is nonempty, compact and controlled invariant. The period in this formula is allowed to vary.

The comparison uses two constructions: a spanning family gives a finite invariant partition, and the words of an invariant partition give spanning families at longer horizons. First, spanning families concatenate, so $r_{\text{inv}}(m+n, Q) \leq r_{\text{inv}}(m, Q)r_{\text{inv}}(n, Q)$ when the right-hand side is finite. Moreover $r_{\text{inv}}(n, Q)$ is nondecreasing. If any finite-horizon spanning number is finite, then $r_{\text{inv}}(1, Q)$ is finite, all of them are finite by concatenation, and Fekete's lemma gives a limit in (2.3). If none is finite, that limit equals $+\infty$.

Let $S_N = \{\omega^1, \dots, \omega^m\}$ be a minimal finite N -step spanning family. Its safe sets are

$$K_i = Q \cap \bigcap_{s=1}^N \{x \in X : \varphi(s, x, \omega^i) \in Q\}.$$

They are closed in Q , since finite control compositions are continuous and Q is closed. They cover Q . Set $D_1 = K_1$ and $D_i = K_i \setminus \bigcup_{j < i} K_j$, discard empty sets, and assign ω^i to D_i . This gives a Borel invariant partition $\mathcal{C}_N$ of period N with at most m atoms, using the spanning construction in Kawan's Theorem 2.3.

For the other inequality in (6.3), all admissible k -block words of any $\mathcal{C}$ form a spanning family after their assigned controls are concatenated. Indeed, from each initial state the feedback rule selects one such admissible word. Consequently $r_{\text{inv}}(k\tau, Q) \leq \#W_k(\mathcal{C}; Q)$, and taking limits gives $h_{\text{inv}}(Q) \leq h(\mathcal{C}; Q)$. The reverse inequality follows from $h(\mathcal{C}_N; Q) \leq N^{-1} \log r_{\text{inv}}(N, Q)$ and $N \rightarrow \infty$. If no finite spanning family exists, no finite invariant partition exists, and both sides are $+\infty$.

Thus, for every nonempty fixed-period subclass $\mathfrak{C} \subset \mathfrak{C}_{\text{Bor}}(Q, \tau)$, the general comparison is

$$h_{\text{inv}}(Q) \leq \inf_{\mathcal{C} \in \mathfrak{C}} h(\mathcal{C}; Q) \leq \inf_{\mathcal{C} \in \mathfrak{C}} h^*(\mathcal{C}; Q). \quad (6.4)$$

The Thue–Morse realization below shows that the last quantity can strictly exceed $h_{\text{inv}}(Q)$.

6.3. Why the unrestricted all-period infimum collapses. Allowing the control period to vary gives the quantity

$$H_{\text{pat,all}}^{*,\text{Bor}}(Q) := \inf_{\tau \in \mathbb{N}} \inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Q, \tau)} h^*(\mathcal{C}; Q).$$

Kawan's spanning construction gives the following direct consequence. The alphabet of an N -step partition already bounds every sampled count, regardless of the observation gaps.

Proposition 6.1 (All-period collapse). *In the full-control discrete-time setting of Section 2, every nonempty compact controlled-invariant set Q satisfies*

$$H_{\text{pat,all}}^{*,\text{Bor}}(Q) = h_{\text{inv}}(Q). \quad (6.5)$$

Proof. The lower bound follows from (6.4). For the upper bound use the same $\mathcal{C}_N$ just constructed. Since its alphabet has at most $r_{\text{inv}}(N, Q)$ symbols, every pattern on k coordinates has at most $r_{\text{inv}}(N, Q)^k$ possibilities. Hence

$$H_{\text{pat,all}}^{*,\text{Bor}}(Q) \leq h^*(\mathcal{C}_N; Q) \leq \frac{\log r_{\text{inv}}(N, Q)}{N}.$$

Let $N \rightarrow \infty$. The infinite-entropy case was already covered by nonexistence of finite invariant partitions. This proves (6.5). $\square$

The all-period identity does not produce a common physical observation sequence. A block index a_i corresponds to time τa_i , which changes with τ . It says only that packing longer control words into one observed symbol can lower the normalized pattern rate.

6.4. Symbolic safety realizations. To transfer a difference between consecutive and sparse symbolic counts to a control problem, we force the safe control to equal the current state symbol. This leaves no choice of physical label, while still allowing arbitrary Borel refinements of its observation.

6.4.1. *Matching the control to the current symbol.* Let E be a finite nonempty discrete alphabet and let $\emptyset \neq Y \subset E^{\mathbb{N}_0}$ be closed with $\sigma(Y) \subset Y$. Write

$$\mathcal{L}_N(Y) = \{y_0 \cdots y_{N-1} : y \in Y\}, \quad P_Y(F) = \#\{y|_F : y \in Y\}.$$

The coordinate complexities are

$$h_{\text{seq,coord}}^A(Y) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log P_Y(\{a_1, \dots, a_n\}),$$

$$h_{\text{pat,coord}}^*(Y) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \max_{\substack{F \subset \mathbb{N}_0 \\ \#F=n}} P_Y(F).$$

Adjoin an isolated cemetery point and give $X = Y \sqcup \{\dagger\}$ the topological disjoint-union topology. Put $U = E$ and define

$$f(y, u) = \begin{cases} \sigma y, & y_0 = u, \\ \dagger, & y_0 \neq u, \end{cases} \quad f(\dagger, u) = \dagger. \quad (6.6)$$

Proposition 6.2 (Symbolic-to-control realization). *The matching-control system (6.6) is continuous on compact metrizable X , and Y is controlled invariant. Its finite-horizon spanning count is, for every $N \in \mathbb{N}$,*

$$r_{\text{inv}}(N, Y) = \#\mathcal{L}_N(Y), \quad (6.7)$$

and therefore

$$h_{\text{inv}}(Y) = h_{\text{top}}(Y, \sigma) = \lim_{N \rightarrow \infty} \frac{1}{N} \log \#\mathcal{L}_N(Y). \quad (6.8)$$

At period one identify U^1 with U . If $\mathcal{C} = (\{D_i : i \in I_{\mathcal{C}}\}, 1, \nu) \in \mathfrak{C}_{\text{Bor}}(Y, 1)$, set $\lambda_{\mathcal{C}}(i) = \nu(D_i) \in E$. Then every atom satisfies

$$D_i \subset [\lambda_{\mathcal{C}}(i)]_Y := \{y \in Y : y_0 = \lambda_{\mathcal{C}}(i)\}. \quad (6.9)$$

Moreover, the one-block label map

$$\lambda_{\mathcal{C}} : \Sigma_{\mathcal{C}} \longrightarrow Y, \quad (a_j)_{j \geq 0} \longmapsto (\lambda_{\mathcal{C}}(a_j))_{j \geq 0}, \quad (6.10)$$

is a continuous, onto, shift-commuting map. Consequently, for every $A \in \mathcal{S}$,

$$\inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Y, 1)} h_{\text{seq}}^A(\mathcal{C}; Y) = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{clon}}(Y, 1)} h_{\text{seq}}^A(\mathcal{C}; Y) = h_{\text{seq,coord}}^A(Y), \quad (6.11)$$

$$\inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Y, 1)} h^*(\mathcal{C}; Y) = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{clon}}(Y, 1)} h^*(\mathcal{C}; Y) = h_{\text{pat,coord}}^*(Y), \quad (6.12)$$

$$\inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Y, 1)} h(\mathcal{C}; Y) = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{clon}}(Y, 1)} h(\mathcal{C}; Y) = h_{\text{top}}(Y, \sigma). \quad (6.13)$$

Every Borel and clopen infimum in (6.11)–(6.13) is attained by the same first-coordinate cylinder partition.

Proof. Safety forces the control word to match the state prefix. This gives the spanning count directly. For a refined partition, we then project atom labels back to those forced controls, obtaining the lower bound for every finite observation window. The first-coordinate partition attains all these bounds.

The space X is the union of the two clopen compact sets Y and $\{\dagger\}$, hence is compact metrizable. For fixed $u \in E$, the sets $[u]_Y$, $Y \setminus [u]_Y$, and $\{\dagger\}$ are clopen; on them f_u is, respectively, the shift or a constant. Thus f_u is continuous. Since U is finite discrete, f is jointly continuous. Choosing $u = y_0$ sends y to $\sigma y \in Y$, proving controlled invariance.

A direct induction gives

$$\varphi(n, y, \omega) \in Y \quad (0 \leq n \leq N) \iff \omega_j = y_j \quad (0 \leq j < N). \quad (6.14)$$

Indeed, after j matches the state is $\sigma^j y$, whereas a mismatch enters $\dagger$ permanently. Hence a length- N control word is safe from at least one state of Y exactly when it belongs to $\mathcal{L}_N(Y)$, and then it safely covers precisely its nonempty prefix cylinder. Distinct language words cover

disjoint cylinders, and every such cylinder is nonempty. A spanning family must therefore contain every word of $\mathcal{L}_N(Y)$; those words also suffice. This proves (6.7). The language counts are submultiplicative, so their normalized logarithms converge.

For completeness, this limit is the usual topological entropy even when $\sigma|_Y$ is not onto. The join of the first-coordinate cylinder cover through times $0, \dots, N-1$ has exactly $\#\mathcal{L}_N(Y)$ nonempty atoms. Conversely, given a finite open cover of compact Y , the partition into cylinders of some fixed length m refines that cover; its N -fold dynamical join is then refined by the length- $(N+m-1)$ cylinders. Taking exponential rates proves (6.8).

Let $\mathcal{C} = (\{D_i : i \in I_{\mathcal{C}}\}, 1, \nu)$ be a Borel invariant partition. If D_i contained y with $y_0 \neq \lambda_{\mathcal{C}}(i)$, its assigned control would send y to $\dagger \notin Y$, contradicting safety. This proves (6.9).

Take $a = (a_j)_{j \geq 0} \in \Sigma_{\mathcal{C}}$ and put $z_j = \lambda_{\mathcal{C}}(a_j)$. For each N , choose a witness $y^{(N)}$ for the first N symbols of a . Induction using matching controls and (6.9) gives $y_j^{(N)} = \lambda_{\mathcal{C}}(a_j) = z_j$ for $0 \leq j < N$. Hence $y^{(N)} \rightarrow z$ in the product topology; since Y is closed, $z \in Y$. Thus $\Lambda_{\mathcal{C}}$ is well defined, and it is continuous: the preimage of a cylinder in $E^{\mathbb{N}_0}$ is a finite union of cylinders in $I_{\mathcal{C}}^{\mathbb{N}_0}$, restricted to $\Sigma_{\mathcal{C}}$. It is shift commuting because $(\Lambda_{\mathcal{C}}(\sigma a))_j = \lambda_{\mathcal{C}}(a_{j+1}) = (\sigma \Lambda_{\mathcal{C}}(a))_j$. Given $y \in Y$, let a_j be the unique index with $\sigma^j y \in D_{a_j}$. Its assigned label is y_j , and y witnesses every finite prefix of this atom itinerary. Thus (6.10) is onto. The factor maps compact name spaces: the limiting point z need not follow the same Borel atom itinerary a . Only its forced control labels are used in the convergence argument.

For every finite F , the coordinate projection of $\Lambda_{\mathcal{C}}$ is onto, so

$$P_{\mathcal{C}}(F) \geq P_Y(F). \quad (6.15)$$

It follows that every Borel, and hence every clopen, invariant partition has sequence, maximal-pattern, and consecutive-name entropy at least the corresponding coordinate quantity.

Let $E_Y = \{e \in E : [e]_Y \neq \emptyset\}$ and define

$$\mathcal{C}_0 = (\{D_e = [e]_Y : e \in E_Y\}, 1, \nu_0), \quad \nu_0(D_e) = e.$$

This is a clopen invariant partition. Under the symbol identification, $W_N(\mathcal{C}_0; Y) = \mathcal{L}_N(Y)$ and $\Sigma_{\mathcal{C}_0} = Y$; the latter follows from closedness of Y . Equality holds in (6.15) for every F . Thus $\mathcal{C}_0$ attains all the lower bounds, proving (6.11)–(6.13). $\square$

6.5. A Thue–Morse separation. Let $\vartheta(0) = 01$, $\vartheta(1) = 10$, and let

$$\mathbf{t} = \lim_{r \rightarrow \infty} \vartheta^r(0) = 0110100110010110 \dots$$

Equivalently, $t_n = s_2(n) \pmod{2}$, where $s_2(n)$ is the number of ones in the binary expansion of n . Let

$$Y = \overline{\{\sigma^m \mathbf{t} : m \geq 0\}} \subset \{0, 1\}^{\mathbb{N}_0}. \quad (6.16)$$

This is the one-sided Thue–Morse orbit closure. The following direct computation supplies both rates used below.

Proposition 6.3 (Exact symbolic rates). *For every $N \geq 1$,*

$$\#\mathcal{L}_N(Y) \leq 8N. \quad (6.17)$$

Hence $h_{\text{top}}(Y) = 0$. On the single schedule

$$A_{\text{TM}} = (a_i)_{i \geq 1}, \quad a_i = 4^{i-1} - 1, \quad (6.18)$$

one has, for every $k \geq 1$,

$$P_Y(\{a_1, \dots, a_k\}) = 2^k. \quad (6.19)$$

The chosen coordinate set contains zero. Since a cylinder is clopen, its patterns in the orbit closure are exactly the patterns obtained along shifts of $\mathbf{t}$; this calculation uses actual shifts of the fixed point. Consequently

$$h_{\text{seq, coord}}^{A_{\text{TM}}}(Y) = h_{\text{pat, coord}}^*(Y) = \log 2. \quad (6.20)$$

Proof. A finite word is in $\mathcal{L}_N(Y)$ exactly when it occurs as a factor of $\mathbf{t}$: one direction follows from the orbit, and the other from convergence in a prefix cylinder. Choose $\ell = \lceil \log_2 N \rceil$. The identity $\mathbf{t} = \vartheta^\ell(\mathbf{t})$ decomposes $\mathbf{t}$ into aligned supertiles of length 2^ℓ . Since $N \leq 2^\ell$, every length- N factor is contained in $\vartheta^\ell(ab)$ for an adjacent two-letter factor ab . There are at most four choices for ab and at most 2^ℓ starting offsets in the first supertile. Therefore $\#\mathcal{L}_N(Y) \leq 4 \cdot 2^\ell \leq 8N$, proving zero topological entropy.

For the sparse lower bound, we encode each prescribed bit at an even binary position. The intervening odd positions stop carries, while one higher bit adjusts the total parity.

Fix $k \geq 1$ and $\varepsilon = (\varepsilon_0, \dots, \varepsilon_{k-1}) \in \{0, 1\}^k$. Put

$$r = \#\{0 \leq i < k : \varepsilon_i = 0\}, \quad q = \sum_{\substack{0 \leq i < k \\ \varepsilon_i = 0}} 4^i + (r \bmod 2)4^k.$$

Only even binary positions occur in this sum. The binary digit sum of q is $r + (r \bmod 2)$, which is even, so $t_q = 0$. If $\varepsilon_i = 1$, bit $2i$ of q is zero, and adding $4^i = 2^{2i}$ changes the digit-sum parity. If $\varepsilon_i = 0$, bit $2i$ is one and bit $2i + 1$ is zero; adding 2^{2i} replaces these two bits by zero and one without a further carry, so the parity is unchanged. In both cases

$$t_{q+4^i} = \varepsilon_i \quad (0 \leq i < k). \quad (6.21)$$

The extra bit at position $2k$ only corrects the initial parity of q . For $i < k$, a carry stops at the empty odd position $2i + 1$, which is strictly below $2k$. The parity correction therefore interferes with none of these tests.

Set $n = q + 1$. Since $a_{i+1} = 4^i - 1$,

$$(\sigma^n \mathbf{t})_{a_{i+1}} = t_{q+4^i} = \varepsilon_i.$$

Thus every binary k -pattern occurs on the first k members of A_{TM} , proving (6.19). The binary alphabet gives the reverse upper bound $p_Y^*(k) \leq 2^k$, and (6.20) follows. $\square$

Corollary 6.4 (Zero invariance entropy, full binary sparse-time complexity). *Apply the matching-control construction (6.6) to the Thue–Morse subshift (6.16). Then*

$$h_{\text{inv}}(Y) = 0 < \log 2 = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Y, 1)} h^*(\mathcal{C}; Y) = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{clon}}(Y, 1)} h^*(\mathcal{C}; Y). \quad (6.22)$$

Moreover, the explicit schedule A_{TM} satisfies

$$\max_{A \in \mathcal{S}} \inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Y, 1)} h_{\text{seq}}^A(\mathcal{C}; Y) = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Y, 1)} h_{\text{seq}}^{A_{\text{TM}}}(\mathcal{C}; Y) = \log 2, \quad (6.23)$$

$$\max_{A \in \mathcal{S}} \inf_{\mathcal{C} \in \mathfrak{C}_{\text{clon}}(Y, 1)} h_{\text{seq}}^A(\mathcal{C}; Y) = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{clon}}(Y, 1)} h_{\text{seq}}^{A_{\text{TM}}}(\mathcal{C}; Y) = \log 2. \quad (6.24)$$

Proof. Combine Proposition 6.2 with Proposition 6.3. For any schedule the binary coordinate complexity is at most $\log 2$, while A_{TM} attains $\log 2$. Equations (6.11)–(6.12) give the Borel and clon infima. $\square$

The separation persists after minimization over all Borel invariant partitions at period one. It therefore distinguishes the fixed-period pattern infimum from invariance entropy.

For the matching-control Thue–Morse example, the collapse is particularly visible. At period N , the length- N cylinder partition with its matching control blocks is clon, has at most $8N$ atoms and has pattern rate at most $N^{-1} \log(8N)$. Thus even the all-period clon infimum is zero in this example, while the period-one Borel infimum is $\log 2$. Changing the control period changes which information is packed into one observed symbol and the cost assigned to that symbol.

7. AFFINE EXAMPLES AND FEEDBACK CONSISTENCY

For expanding interval branches, a prescribed word can be followed only from a small set of initial states. Covering the interval therefore requires many words. The first example matches this volume bound by a feedback that eventually enters an invariant core. The second uses protected inverse branches to force every sampled word under every feedback. The four-branch example then separates what these arguments can establish: its maximal-pattern minimum is $\log 2$, while its common-sampling value is still only bounded. We finish by showing why an efficient open-loop covering need not define a feedback.

7.1. Integer affine branches and an absorbing feedback. Let $m \geq 2$, $D \geq m$ be integers, $R = D/(m-1)$, $Q = [0, R]$, and controls $u = 0, \dots, D$ act legally by $mx - u$. On $X = [-1, R+1]$ define the continuous ambient maps

$$f_u(x) = \max\{-1, \min\{R+1, mx - u\}\}.$$

For $x \in Q$, this is in Q exactly when $mx - u$ is in Q , and then truncation has no effect. The safety intervals $K_u = [u/m, (u+R)/m]$ cover Q . Fix any period $\tau \geq 1$.

Proposition 7.1. *For this system, over all legal Borel block feedbacks and finite Borel atom refinements,*

$$\min_s h_{(0,1,\dots)}(s) = L_\tau = M_\tau = \log m.$$

There is an actual Borel selector s_τ with $h_A(s_\tau) = h^(s_\tau) = \log m$ for every infinite increasing A , although its absorption time has no uniform bound.*

Proof. Expansion gives the lower bound by making each word's initial-state fiber small. For the upper bound we construct a rule under which every state except the right endpoint eventually enters the usual m -digit core. Entry times are not uniformly bounded, so we count where the transition can occur among the observed times, rather than bounding the elapsed waiting time.

For any actual length- n block word, concatenate its controls into $w_0 \cdots w_{n\tau-1}$. Safety gives

$$x_{n\tau} = m^{n\tau}x - \sum_{j=0}^{n\tau-1} w_j m^{n\tau-1-j} \in [0, R].$$

Its initial-state fiber is contained in an interval of length $R/m^{n\tau}$, whether or not that fiber itself is an interval. These containing intervals cover Q , so there are at least $m^{n\tau}$ actual consecutive block words. This gives the full-class lower bound at the common sequence $(0, 1, \dots)$.

Define the one-step rule $v(x) = \min\{D, \lfloor mx \rfloor\}$ and $T(x) = mx - v(x)$. If $x < D/m$, the next state is in $[0, 1)$; if $x \geq D/m$, it is $mx - D \in [0, R]$. The core $[0, 1)$ is invariant and uses only $\Gamma = \{0, \dots, m-1\}$. Both 0 and R are fixed; D/m uses D and goes to zero. For $x < R$, while D is used the state is $R - m^j(R - x)$. Since $R - x > 0$, eventually it is below D/m , then one transitional control $v < D$ sends it into the core. Only R uses D forever. The absorption times are unbounded as x tends to R .

From the current state define $s_\tau(x) = (v(x), v(Tx), \dots, v(T^{\tau-1}x))$. It is Borel, each intermediate step is safe, and its actual endpoint map is T^τ . Put $\mathcal{B} = \Gamma^\tau$, $b = m^\tau$, and $d_0 = (D, \dots, D)$. Its block names are contained in

$$\bigcup_{j \geq 0} d_0^j \mathcal{C}_\tau \mathcal{B}^{\mathbb{N}_0} \cup \{d_0^\infty\}, \quad |\mathcal{C}_\tau| \leq D \frac{m^\tau - 1}{m - 1} =: C_\tau.$$

The transitional block has some initial D 's, one control below D , and then core controls; this proves the stated count. For any window of n block positions, an unobserved transition leaves just one of $n+1$ cuts between constant initial labels and core labels. A sampled transition has n possible locations and at most C_τ labels. Thus

$$\mathbf{P}_{s_\tau}(F) \leq [n+1 + nC_\tau]m^{\tau n}, \quad |F| = n.$$

There is no factor depending on the last observation time or on the unbounded absorption time.

For the matching lower count prescribe any n core blocks at the positions of F , fill all other one-step digits with zero, and let $x = \sum_{j \geq 0} e_j m^{-j-1} < 1$. There are only finitely many nonzero digits, so each remaining base- m tail is smaller than one. Induction gives $\lfloor mT^j x \rfloor = e_j$ and shows that the rule generates the prescribed digits. Hence $P_{s_\tau}(F) \geq m^{\tau n}$ for every window. The all-window bounds and the full-class consecutive lower bound prove the proposition, with minima actually attained. Control-label projection treats arbitrary atom refinements. $\square$

When $m \geq 3$ and $D = m$, the minimum one-step safe subcover has $r = m + 1$, although $L_1 = M_1 = \log m$. Indeed, m intervals of length R/m could cover an interval of length R only by abutting without positive-length overlap. Their left endpoints are multiples of $1/m$, so this would force their consecutive control differences to equal the noninteger $R = m/(m - 1)$. All $m + 1$ intervals are therefore needed. Although the legal branches have different successors, they share the state factor $x \mapsto e^{2\pi i x}$, with induced map $z \mapsto z^m$.

7.2. Protected inverse branches and common sampling. On $Q = [0, 1]$, at period one, use

$$g_a(x) = \frac{5}{4}x, \quad g_b(x) = \frac{20}{3}x - \frac{16}{3}, \quad g_c(x) = 20x - 19, \quad g_d(x) = 100x.$$

Their safety domains are $[0, 4/5]$, $[4/5, 19/20]$, $[19/20, 1]$, and $[0, 1/100]$, respectively. Continuous ambient extensions are obtained by truncating these affine maps to $[-1, 2]$, with no change on legal paths. The minimum subcover is three and the largest two inverse slopes sum to $4/5 + 3/20 = 19/20 < 1$; hence $M_1 = \log 3$ by the scalar special case of the capacity theorem.

Proposition 7.2. *For $A_2 = (0, 2, 4, \dots)$, every legal Borel feedback and every finite Borel atom refinement has at least 3^n actual patterns on its first n observations. Consequently*

$$\min_s h_{A_2}(s) = L_1 = M_1 = \log 3.$$

Proof. We construct two-step inverse paths whose first controls can be chosen freely and whose second control is always b . Each path lies in regions where safety forces those controls, so the construction works for every feedback on the same even sampling sequence.

The full inverse branches are $\psi_a(y) = 4y/5$, $\psi_b(y) = 4/5 + 3y/20$, $\psi_c(y) = 19/20 + y/20$. Let $J = [1/2, 999/1000]$ and $\theta_u = \psi_u \psi_b$. They satisfy

$$\theta_a(y) = 16/25 + 3y/25, \quad \theta_b(y) = 23/25 + 9y/400, \quad \theta_c(y) = 99/100 + 3y/400.$$

Their complete images and the intermediate image are

image	exact endpoints	only legal control
$\psi_b(J)$	$[7/8, 18997/20000]$	b
$\theta_a(J)$	$[7/10, 18997/25000]$	a
$\theta_b(J)$	$[149/160, 376991/400000]$	b
$\theta_c(J)$	$[159/160, 398997/400000]$	c .

Each $\theta_u(J)$ lies in J . Every displayed closed interval is strictly inside the appropriate uniquely legal region; in particular $\theta_a(J)$ avoids the extra control d .

Given any $u_0 \cdots u_{n-1} \in \{a, b, c\}^n$, set $x_{2n} = 1/2$ and recursively $x_{2j+1} = \psi_b(x_{2j+2})$, $x_{2j} = \theta_{u_j}(x_{2j+2})$. The table shows that at every even state only u_j is legal and at every odd state only b is legal. Thus *every* feedback must execute the actual period-one prefix $u_0, b, \dots, u_{n-1}, b$ from the same constructed initial state. The endpoint remains in Q and the globally legal rule continues safely forever. There is no added phase or period-two control. All three-symbol sampled words are therefore actual patterns. Refinement only increases their count.

The Borel selector using a below $4/5$, b from $4/5$ to $19/20$, and c at and above $19/20$, is legal at all endpoints and uses only three labels. Its upper count is 3^n on every window. This matches the lower bound, including attainment, and $L_1 \leq M_1$ completes the equalities. The normalization is $n\tau = n$, not $2n$. $\square$

The closed contracting branches $\theta_a, \theta_b, \theta_c$ have disjoint first-level images in J . Their nested images give a compact set homeomorphic to the three-symbol full shift, and every T_s^2 on that set is the same shift under this coding. To verify this, a word's image has diameter at most $(3/25)^n \text{diam } J$; nested images intersect in one point, different first differing letters give disjoint images, and the inverse identities used above give the shift relation. The union of this set with its ψ_b image is protected under every legal feedback. These sets can overlap: the all- b state $16/17$ belongs to both and uniquely uses b . Hence the proof does not introduce an implicit disjoint phase register. This is precisely a protected-subsystem mechanism, even if the rest of the state space has feedback-dependent successors.

7.3. A noninteger four-branch system. Let $Q = [0, 6]$ and use, at period one,

$$g_{A0}(x) = \frac{3}{2}x, \quad g_{A1}(x) = \frac{3}{2}x - 3, \quad g_{B0}(x) = \frac{5}{3}x, \quad g_{B1}(x) = \frac{5}{3}x - 4.$$

The original safety intervals are $[0, 4]$, $[2, 6]$, $[0, 18/5]$, $[12/5, 6]$. Truncation to the ambient interval $[-1, 7]$ retains exactly these safe transitions. All legal Borel feedbacks, arbitrary mixtures, and finite atom refinements are allowed. Two A domains cover Q and no single one does, so $r = 2$. The largest scalar inverse capacity is $2/3 < 1$. Thus $M_1 = \log 2$. Every actual length- N control prefix has its fiber inside an interval of length at most $6(3/2)^{-N}$; these intervals cover Q . Its number $D_N(s)$ is therefore at least $(3/2)^N$. Appendix E retains the weights and actual even-odd compatibility in a refinement of this volume argument. This gives

$$\log(3/2) \leq L_1 \leq \log 2. \quad (7.1)$$

For $A_2 = (0, 2, \dots)$ let $\ell_2 = \inf_s h_{A_2}(s)$, and let ℓ_2^A restrict to all two- A Borel feedbacks, without restricting their partitions. The unrefined two- A rules have the exact description

$$s_E(x) = A0 \text{ on } [0, 2) \cup E, \quad s_E(x) = A1 \text{ elsewhere}, \quad E \subset [2, 4] \text{ Borel}.$$

Indeed the two overlap domains are exactly $[2, 4]$. Write $N_n(s) = P_s(\{0, 2, \dots, 2n - 2\})$ for the even pattern count. For any feedback, an actual $2n$ prefix is determined by its even and odd sampled patterns. Odd patterns occur among the even patterns from states $T_s x \in Q$. Hence $D_{2n}(s) \leq N_n(s)^2$ and the volume lower bound gives $N_n(s) \geq (3/2)^n$. Consequently $\ell_2 \geq \log(3/2)$.

Proposition 7.3 (Density-one samplings). *If $A = (a_i)$ satisfies $a_n/n \rightarrow 1$, then $\inf_s h_A(s) = \log(3/2)$ over the full class. The greedy feedback $s_g = s_\varnothing$ is an actual minimizer for every such A .*

Proof. The consecutive counts of the greedy rule grow like $(3/2)^n$. To establish the sampled statement, we first bound these counts on the entire interval, including points that have not yet reached the invariant core. A density-one sequence omits only $o(n)$ coordinates from its elapsed prefix, and that omission changes the logarithmic count by $o(n)$.

Set $\beta = 3/2$. On the invariant core $[0, 3)$, scale $x = 3t$. The greedy dynamics becomes $B(t) = \beta t - \lfloor \beta t \rfloor$ on $[0, 1)$. Every nonempty length- n digit cylinder is a left-closed right-open interval whose B^n image is $[0, l)$ for some $0 < l \leq 1$. This is proved by splitting the current image at $1/\beta$: it has a single child if $\beta l \leq 1$, and otherwise a full-image child and a nonfull-image child. Equality creates no extra singleton at the excluded right endpoint.

Let F_n be the number of full-image cylinders, with $F_0 = 1$. The sum of image lengths is β^n , since each source length is l/β^n and the sources partition $[0, 1)$. Thus $F_n \leq \beta^n$. Each nonfull cylinder has a last full-image ancestor, and each ancestor has at most one descendant

at a given later level without another full-image step. The total core count C_n consequently satisfies

$$C_n \leq \sum_{j=0}^n F_j \leq 3\beta^n.$$

For $x \in [3, 6)$ the greedy rule uses A_1 until entering the core; while waiting its state is $6 - \beta^j(6 - x)$, so entry is finite. State 6 uses A_1 forever. Therefore the full-space count satisfies

$$D_n(s_g) \leq 1 + \sum_{j=1}^n C_j \leq 10\beta^n.$$

This includes all thresholds and the exceptional endpoint.

Let $N = a_n + 1$, so $N - n = o(n)$. Each sampled word has at most 4^{N-n} completions to a consecutive word, giving

$$P_s(\{a_1, \dots, a_n\}) \geq \frac{D_N(s)}{4^{N-n}} \geq \frac{\beta^N}{4^{N-n}}.$$

For the greedy rule the same count is at most $D_N(s_g) \leq 10\beta^N$. After division by n the two rates tend to $\log \beta$. Refinements preserve the lower bound and the unrefined rule gives the matching upper bound. $\square$

Let $\rho > 1$ be the unique solution

$$\rho^6 = \rho^5 + \rho^4 + \rho + 1, \quad \log \rho = 0.55600938749 \dots \quad (7.2)$$

Appendix D computes the greedy feedback's exact rate $h_{A_2}(s_g) = \log \rho$, with full upper and lower counts. Combining these estimates gives

$$\log(3/2) \leq \ell_2 \leq \ell_2^A \leq \log \rho. \quad (7.3)$$

Thus we know the greedy rate at this one sequence, but not whether it minimizes that rate among two- A rules or among all four-control rules. Optimizing the sequence as well gives the separate bounds (7.1). The consistency obstruction below explains why an open-loop language with favorable counts is not enough to improve either upper bound.

7.4. An alternating open-loop language with no Borel realization. Let $R > 0$, $1 < a, b < 2$, and $\log a / \log b \notin \mathbb{Q}$. On $[0, R]$ take four legal affine branches

$$g_{Ai}(x) = ax - i(a - 1)R, \quad g_{Bi}(x) = bx - i(b - 1)R \quad (i = 0, 1).$$

Each type's two full inverse intervals cover Q . Let Λ_{alt} be the two-phase language whose A, B types alternate strictly and whose digits are otherwise arbitrary. Let

$$\widehat{\Lambda} = \Lambda_{\text{alt}} \cup \{wc^\infty : w \text{ is a finite alternating-type word, } c \in U\}.$$

The empty prefix is allowed. This compact shift-invariant language includes all constant endpoint tails and their finite alternating prehistories. The inverse contractions $\psi_{Ci}(y) = (y + i(\lambda_C - 1)R)/\lambda_C$, with $\lambda_A = a, \lambda_B = b$, define a continuous coding $\pi(d) = \lim_n \psi_{d_0} \cdots \psi_{d_{n-1}}(0)$. Every state has a coding in Λ_{alt} : in each prescribed type choose a safe digit and repeat. The contraction error is at most $R \min(a, b)^{-n}$. This proves full open-loop coverage, not feedback consistency.

Proposition 7.4. *No globally safe Borel period-one feedback has all its control names in $\widehat{\Lambda}$. This remains true with arbitrary Borel mixtures and finite atom refinements. Nevertheless for every n -position window F with $n \geq 1$,*

$$2^{n+1} \leq P_{\widehat{\Lambda}}(F) \leq 8(n+1)2^n.$$

Proof. The alternating part has two distinct type patterns and 2^n independent digits. For a constant-tail member there are at most $n + 1$ cuts on F , two prefix phases, 2^j prefix digits, and four constant tails. Summing by cuts bounds the count by $8(n + 1)2^n$ (and also covers the infinite alternating case).

To prove nonrealizability, examine small positive states. There only zero-digit controls are safe, so two alternating steps multiply the state by ab . In logarithmic coordinates, a Borel choice of the first type would give a two-coloring of a circle exchanged by an irrational rotation. The following argument rules out that coloring.

Let C be the union of all finite affine preimages of $\{0, R\}$. This set is countable, and its complement is preserved by every legal branch: a successor in C would put the original state in C . A constant safe tail under any one branch forces its tail state to be the corresponding endpoint, by expansion about 0 or R . Thus any hypothetical feedback has strictly alternating types forever on $Q \setminus C$. Choose

$$0 < \eta < \min\{R/a, R/b, (a-1)R/a, (b-1)R/b\}.$$

On $(0, \eta)$ only zero-digit controls are legal. For $0 < x < \eta/(ab)$ outside C , the first two steps are x, ax, abx if the first type is A , and x, bx, abx otherwise. The third type is the first type. Put $p = \log a, q = \log b, h = p + q$ and encode the feedback type at ηe^t , $t < 0$, by a Borel $c(t) \in \{0, 1\}$, with A represented by one. Outside a countable exceptional set,

$$c(t+h) = c(t), \quad c(t) = 1 \Rightarrow c(t+p) = 0, \quad c(t) = 0 \Rightarrow c(t+q) = 1 \quad (t < -h).$$

Translating each negative interval of length h to $[-h, 0)$ gives a measurable circle function, with all countably many translated exceptions still null. Apply the last two implications at $t = r - h$, $-h \leq r < 0$. For its set E of type one, circle rotations satisfy $R_p E \subset E^c$ and $R_q E^c \subset E$ modulo null sets. Since $q = -p$ modulo h , these imply $R_p E = E^c$, measure $1/2$, and $R_{2p} E = E$ modulo null sets. The number $2p/h$ is irrational. For $G = 2\mathbf{1}_E - 1$, invariance under that rotation and Fourier coefficients give $\widehat{G}(j) = e^{2\pi i j(2p/h)} \widehat{G}(j)$. Every nonzero coefficient vanishes; the mean vanishes as well. Completeness of the trigonometric system gives $G = 0$ almost everywhere, contradicting $|G| = 1$. This proves impossibility. Refinements project to the same physical control rule and cannot avoid the obstruction. $\square$

The noninteger system in Section 7.3 has $a = 3/2, b = 5/3, R = 6$. Its log ratio is irrational by unique prime factorization: $(3/2)^j = (5/3)^k$ would imply $3^{j+k} = 2^j 5^k$. None of its nonempty finite affine composites has an integer slope. Thus its alternating A/B open-loop language cannot be realized by a globally safe Borel feedback. Other feedbacks remain available, so this obstruction does not determine the optimized sampling rates.

8. THE REMAINING COMMON-SAMPLING PROBLEM

The unresolved step in the capacity approach is the choice of times. The product estimate forces independent coordinates for each feedback, but those coordinates may move when the rule changes. Does it nevertheless imply $L_\tau = M_\tau$ for the full Borel feedback class, or can the two values differ?

The four-branch affine system makes this question concrete:

$$\log(3/2) \leq L_1 \leq M_1 = \log 2.$$

At density-one samplings the inner minimum is $\log(3/2)$; at even times the greedy rate is $\log \rho$, with ρ given by (7.2). Neither the optimal even-time rate nor L_1 is determined. The weighted pattern-pair identity in Appendix E identifies possible improvements to the volume lower bound, but gives no uniform positive improvement. A resolution needs additional information about compatible feedback fibers, beyond an open-loop covering count.

APPENDIX A. ATTAINMENT OF THE FIXED-PERIOD INFIMUM

Integer-logarithm quantization of maximal-pattern entropy is a known combinatorial fact. For one-sided subshifts it is stated in Huang and Ye [4, Theorem 5.5]. We record the finite-set argument needed here, since it also settles attainment of the feedback infimum. This argument does not address the common-sampling problem.

Proposition A.1. *Let $\emptyset \neq \Omega \subset E^{\mathbb{N}_0}$, where $|E| = q < \infty$, and fix $\tau > 0$. There is an integer $m \in \{1, \dots, q\}$ such that $h^*(\Omega; \tau) = \log m / \tau$.*

Proof. For $H \subset E^n$, let $d_k(H)$ be the largest size of a coordinate set J for which $H|_J$ contains $\prod_{j \in J} V_j$, with $V_j \subset E$ and $|V_j| = k$. Put $d_k(\emptyset) = -1$. The elementary finite bound is

$$|H| \leq B_k(n, d) := \sum_{i=0}^{\min(d, n)} \binom{n}{i} (k-1)^{n-i} \binom{q}{k}^i \quad \text{if } d_k(H) \leq d, \quad 2 \leq k \leq q.$$

Set $B_k(n, -1) = 0$. To prove the bound, write H_e for the $(n-1)$ -letter prefixes ending in e , $V = \bigcup_e H_e$, and $V_S = \bigcap_{e \in S} H_e$ for $|S| = k$. Projection gives $d_k(V) \leq d$, whereas $d_k(V_S) \leq d-1$: a shattered set in V_S can be extended by the last coordinate. A prefix appearing in t of the sets H_e contributes $t \leq (k-1) + \binom{t}{k}$. Hence

$$|H| \leq (k-1)|V| + \sum_{|S|=k} |V_S| \leq (k-1)B_k(n-1, d) + \binom{q}{k} B_k(n-1, d-1) = B_k(n, d).$$

The empty class and the nonempty class at $n = 0$ start the induction.

Let $D_k = \sup_F d_k(\Omega|_F)$, over finite $F \subset \mathbb{N}_0$, and let m be the largest k with $D_k = \infty$. Since Ω is nonempty, $D_1 = \infty$. Restricting shattered sets gives $p_\Omega^*(n) \geq m^n$ for every n . If $m = q$, the alphabet bound gives equality. Otherwise $D_{m+1} = d < \infty$, so the preceding bound gives, for a constant C ,

$$m^n \leq p_\Omega^*(n) \leq \sum_{i=0}^{\min(d, n)} \binom{n}{i} m^{n-i} \binom{q}{m+1}^i \leq C n^d m^n \quad (n \geq 1).$$

Taking logarithms and dividing by $n\tau$ proves the assertion. $\square$

Corollary A.2. *For every nonempty class of finite Borel invariant partitions at one fixed period τ , the infimum of $h^*(\mathcal{C}; Q)$ is a minimum. At the common attaining schedule of Corollary 5.6, the inner sequence-entropy infimum is also a minimum. In particular, for a finite control alphabet the quantity M_τ in (2.2) is attained without a countability assumption.*

Proof. Apply Proposition A.1 to each $\Sigma_{\mathcal{C}}$. The corresponding integers form a nonempty subset of $\mathbb{N}$, which has a least member. At the schedule of Corollary 5.6, all individual sequence rates equal their pattern rates. For the full finite-control class, Lemma 2.1 identifies the infimum with that over legal selectors, whose nonempty fibers have at most $|U|^\tau$ labels. $\square$

No attainment assertion for an arbitrary fixed sequence A follows from this corollary. The integer quantization applies to maximal-pattern entropy, and fixed-sequence entropy need not have that form.

APPENDIX B. A QUANTITATIVE SINGLE-STEP ESTIMATE

Assume the notation of Section 4 and $\Theta_k < 1$. The proof below gives an explicit lower density of independent coordinates without the omitted-letter covering lemma. For nonempty $F \subset W^n$, let $d_k(F)$ be the largest $|J|$ for which there are sets $V_j \subset W$, $|V_j| = k$, such that $F|_J \supset \prod_{j \in J} V_j$. The sets V_j may differ with j . Set $d_k(\emptyset) = -1$. Put $\theta = \Theta_k < 1$, $C_0 = \|\sum_u \rho_u\|_\infty$ and $B = C_0 \binom{q}{k}$. For nonnegative f let $\Phi f = \max_{|S|=k-1} \sum_{u \in S} L_u f$ and $\mathcal{T}f = \sum_u L_u f$. Their norms are bounded by θ and C_0 , respectively.

Proposition B.1 (Weighted shattering estimate). *With $H(n, -1) = 0$, for every $F \subset W^n$,*

$$\|P_F\|_{\infty \rightarrow \infty} \leq H(n, d_k(F)), \quad H(n, d) = \sum_{j=0}^{\min(d, n)} \binom{n}{j} B^j \theta^{n-j}. \quad (\text{B.1})$$

For actual names this gives $d_k(F_n(s)) > \alpha n$, uniformly in every legal feedback, whenever $0 < \alpha < 1/2$ satisfies

$$\log \theta + \mathfrak{h}(\alpha) + \alpha \log(B/\theta) < 0, \quad \mathfrak{h}(t) = -t \log t - (1-t) \log(1-t). \quad (\text{B.2})$$

Proof. Take $H(n, -1) = 0$ and treat the empty word at $n = 0$ separately. Split F at its *first* letter: $F_a = \{v : av \in F\}$, $U_0 = \bigcup_a F_a$, and $I_S = \bigcap_{a \in S} F_a$ for $|S| = k$. If $d = d_k(F)$, then $d_k(U_0) \leq d$ and $d_k(I_S) \leq d - 1$: independence of d tail positions in the latter would also make the first position independent and give $d + 1$. For each $v \in U_0$ the set of its permitted first letters has either at most $k - 1$ members, or contains a k -set S . Applying the positive tail product L_v and summing gives

$$P_F f \leq P_{U_0} \Phi f + \sum_{|S|=k} P_{I_S} \mathcal{T} f.$$

This respects the order $L_{av} = L_v L_a$; no operators are commuted. Induction and Pascal's identity give $\theta H(n - 1, d) + B H(n - 1, d - 1) = H(n, d)$. Positive domination handles signed f and proves (B.1).

By actual-fiber mass, $1 \leq H(n, d_k(F_n(s)))$. The cover property gives $C_0 \geq 1$, and positivity gives $0 < \theta \leq C_0 \leq B$. For $d \leq \alpha n$, therefore, $B/\theta \geq 1$ and $\sum_{j \leq \alpha n} \binom{n}{j} \leq e^{n\mathfrak{h}(\alpha)}$ give $H(n, d) \leq \exp(n[\log \theta + \mathfrak{h}(\alpha) + \alpha \log(B/\theta)]) < 1$, a contradiction. The binomial bound follows by comparing each term to its probability under independent Bernoulli(α) trials; those probabilities sum to at most one. Condition (B.2) is possible because its last two terms tend to zero as α tends to zero. $\square$

B.1. A critical boundary of the single-step density condition. For $3/2 < \lambda \leq 2$, on $Q = [0, 1]$ use

$$g_a(x) = \lambda x, \quad g_b(x) = \lambda x - (\lambda - 1)/2, \quad g_c(x) = 3x - 2.$$

The safety intervals are

$$K_a = [0, 1/\lambda], \quad K_b = [(\lambda - 1)/(2\lambda), (\lambda + 1)/(2\lambda)], \quad K_c = [2/3, 1].$$

They cover Q and $r = 3$. Both a and c are forced into every subcover by endpoints $0, 1$. Since $1/\lambda < 2/3$, they leave an open gap, so b is also needed. The inverse densities are constants $1/\lambda, 1/\lambda, 1/3$; therefore $\Theta_2 = 1/\lambda < 1$ and $\Theta_3 = 2/\lambda$.

Proposition B.2. *For every $3/2 < \lambda \leq 2$, this full Borel class has $M_1 = \log 2 < \log 3$. Thus the condition $\Theta_3 < 1$ cannot be uniformly replaced by $\Theta_3 \leq 1$ or by $\Theta_3 < 1 + \eta$ for any $\eta > 0$, while retaining the conclusion $M_1 = \log 3$.*

Proof. The $k = 2$ single-step criterion gives every feedback $h^* \geq \log 2$. Put $t = (\lambda - 1)/(2\lambda)$ and define

$$s_\lambda(x) = \begin{cases} a, & 0 \leq x < t, \\ b, & t \leq x < 2/3, \\ c, & 2/3 \leq x \leq 1. \end{cases}$$

It is Borel and legal: $t < 1/\lambda$ and $(\lambda + 1)/(2\lambda) \geq 3/4 > 2/3$. For $x < t$, $g_a(x) < (\lambda - 1)/2 \leq 1/2$. For $t \leq x \leq 1/2$, $g_b(x) = \lambda x - (\lambda - 1)/2 \in [0, 1/2]$. Hence $J = [0, 1/2]$ is invariant. Before entering J only b, c occur; once in J only a, b occur. This remains true for states which never enter and at all thresholds, including the fixed points $1/2$ and 1 .

At any n selected times a sampled name has some cut $0 \leq j \leq n$ between the two groups, each of size two. Thus its pattern count is at most $(n + 1)2^n$ for *every* window. This actual feedback gives $h^* \leq \log 2$ and completes the equality. At $\lambda = 2$, $\Theta_3 = 1$; as λ increases to 2

from below, Θ_3 decreases to 1 from above. These values give both claimed failures of uniform relaxation. The example does not make $\Theta_3 < 1$ a necessary condition for every system, and does not determine L_1 for this family. $\square$

APPENDIX C. FINITE-CYLINDER REPRESENTATION AND A CONTRACTION CRITERION

The matrix calculation in Section 4.7 works because all iterates of the constant function lie in a finite-dimensional space. We give that reduction for cylinder safety domains, then one criterion that avoids checking products individually. A common positive left weight bounds every list product by the same geometric sequence. This is the usual linear copositive Lyapunov argument for positive matrices; see [1] for the broader framework. The condition below specifies when the safety relations supply such a weight.

C.1. An exact finite representation. Let D be a finite alphabet of size $d \geq 2$, $Q = D^{\mathbb{N}_0}$, and μ its fair Bernoulli probability. The finite controls have legal maps $g_u = P_u \sigma$, where p_u permutes D and P_u acts in every coordinate. Suppose

$$K_u = \bigcup_{\xi \in R_u} [\xi], \quad R_u \subset D^h, \quad h \geq 2,$$

and the K_u cover Q . Shorter cylinders can be completed to the same length h . The continuous cemetery extension used above applies to these clopen domains. The control period is always one.

For every bounded Borel function the full operator is

$$L_u f(y) = \frac{1}{d} \sum_{a \in D} \mathbf{1}_{R_u}(a, p_u^{-1}y_0, \dots, p_u^{-1}y_{h-2}) f(a P_u^{-1}y). \quad (\text{C.1})$$

Indeed, conditioned on $x_0 = a$ the tail of x still has law μ , and P_u preserves that law. Substitution in (4.2) gives this formula with the exact safety gate.

Index source and target vectors by D^{h-1} . For target η and source ξ , set

$$T_u(\eta, \xi) = \frac{1}{d} \# \left\{ a \in D : \begin{array}{l} (a, p_u^{-1}\eta_0, \dots, p_u^{-1}\eta_{h-2}) \in R_u, \\ (a, p_u^{-1}\eta_0, \dots, p_u^{-1}\eta_{h-3}) = \xi \end{array} \right\}. \quad (\text{C.2})$$

When $h = 2$ the second tuple is just a . Functions of the first $h - 1$ coordinates form an invariant subspace of every L_u , represented by T_u . Starting at $\mathbf{1}$, every list product stays in this space. Every such cylinder has positive measure and a positive product has norm equal to its value on $\mathbf{1}$. Therefore, for every length and every history,

$$\|A_{S_{n-1}} \cdots A_{S_0}\|_{\infty \rightarrow \infty} = \|T_{S_{n-1}} \cdots T_{S_0} \mathbf{1}\|_{\infty}, \quad T_S = \sum_{u \in S} T_u. \quad (\text{C.3})$$

Only the operator calculation has become finite-dimensional; the feedback itself may still depend on the whole state.

For two-letter domains, rows are target letters and columns are source letters, with

$$T_u(j, i) = d^{-1} \mathbf{1}_{R_u}(i, p_u^{-1}j), \quad d(w^\top T_u)_i = \sum_{z: (i, z) \in R_u} w(p_u z). \quad (\text{C.4})$$

The structural conditions below use this source-column identity. A left budget $w^\top T_S \leq \theta w^\top$ is different from changing a reference density, which leads instead to a right inequality $T_S a \leq \theta a$. They cannot be interchanged.

C.2. A strict source-weight criterion. Partition $D = A \sqcup B \sqcup C$ into nonempty sets of sizes a, b, c , so $d = a + b + c$. Use three controls with

$$p_0(C) \subset D \setminus A, \quad p_1(B) \subset A, \quad p_2(A) \subset D \setminus A. \quad (\text{C.5})$$

Choose arbitrary relations $E_0 \subset A \times C$ and $E_1 \subset A \times B$, with $E_0 \cup E_1 \neq \emptyset$, and define

$$\begin{aligned} R_0 &= ((D \setminus A) \times C) \cup E_0, \\ R_1 &= ((D \setminus A) \times B) \cup E_1, \\ R_2 &= (A \times D) \cup (D \times A), \quad K_u = [R_u]. \end{aligned} \quad (\text{C.6})$$

If the source letter is in A , control 2 permits every second letter, while E_0 and E_1 specify the extra choices of controls 0 and 1. Outside A , the second letter determines the unique legal control. For $i \in A$ and $u = 0, 1$, write

$$e_u(i) = |\{z : (i, z) \in E_u\}|, \quad e = \max_{i \in A} e_1(i).$$

The following inequality will allow a single choice of weights to control all two-label lists:

$$s_- := \max \left\{ \frac{c}{b+c}, \frac{e}{b+c-e} \right\} < s_+ := \frac{\min\{a, c\}}{b}. \quad (\text{C.7})$$

Its lower bound limits the extra choices on A ; its upper bound limits the weight transferred from sources outside A .

Proposition C.1 (Strict source budget). *For every system satisfying (C.5)–(C.7), $r = 3$, all histories of global two-label lists have exponentially decaying inverse capacity, and every legal unrefined Borel feedback has $h^*(s) = M_1 = \log 3$. Every finite Borel atom refinement has $h^* \geq \log 3$. Moreover, no full-support reference probability can directly satisfy the joint-density or scalar single-step criterion for $k = 3$ on these original branches.*

Proof. We seek one weighted sum of the source coordinates that contracts under every list. Giving the class A a larger weight compensates for its extra legal edges. The interval (s_-, s_+) is precisely what leaves room to choose that weight.

Let $s > 0$ be chosen below, put $\gamma = 1 + s$, and give weight $w_i = \gamma$ to $i \in A$ and weight $w_i = 1$ to the other letters. The weighted mass of the constant vector is $H = \sum_i w_i = d + sa$. Before division by d , the weighted contributions of controls 0, 1, 2 in a source column are

	0	1	2
$i \in A$	$e_0(i) \leq c$	$\gamma e_1(i) \leq \gamma e$	H
$i \notin A$	c	γb	a .

Each entry follows from the original domains and (C.5). For example, on A control 2 permits every tail, hence the sum of all weights irrespective of its permutation. Off A , its permitted tails lie in A and land outside A , each with weight one.

For a source outside A , the two nontrivial requirements that each pair in the second row have sum less than d are

$$c + \gamma b < d, \quad \gamma b + a < d, \quad \text{or equivalently} \quad sb < \min\{a, c\}.$$

For a source in A , compare a pair containing the contribution H with the available weight $d\gamma$. Since $d\gamma - H = s(b + c)$, it is enough to require

$$c < s(b + c), \quad \gamma e < s(b + c).$$

These are exactly $s > s_-$, while the preceding upper restriction is $s < s_+$. Condition (C.7) therefore permits $s = (s_- + s_+)/2$. With this choice $c < H$ and $\gamma e \leq \gamma b < H$, so the pair excluding H is also bounded by $H + \max\{c, \gamma e\}$. Define

$$\theta = \max \left\{ \frac{H + \max\{c, \gamma e\}}{d\gamma}, \frac{c + \gamma b}{d}, \frac{c + a}{d}, \frac{\gamma b + a}{d} \right\}.$$

The inequalities just established, together with $a + c < d$, give $0 < \theta < 1$ and $w^\top T_S \leq \theta w^\top$ for every global two-label list. Repeatedly applying this single inequality gives

$$\|T_{S_{n-1}} \cdots T_{S_0} \mathbf{1}\|_\infty \leq w^\top T_{S_{n-1}} \cdots T_{S_0} \mathbf{1} \leq H\theta^n,$$

since $\min_i w_i = 1$. Equation (C.3) controls the complete capacity for all lengths.

If $x_0 \in A$, control 2 is legal; otherwise its next letter in A, B, C requires, respectively, 2, 1, 0. Thus the domains cover Q . Taking $x_0 \notin A$ and x_1 successively in C, B, A gives cylinders on which each of 0, 1, 2 alone is legal, so $r = 3$. A clopen attaining selector chooses 2 when $x_0 \in A$ or $x_1 \in A$, otherwise 1 when $x_1 \in B$, otherwise 0. It is safe, single-valued, and stays in Q at each step; the same rule at every successor gives all-time safety. Applying Theorem 4.1 proves all pattern conclusions, including the lower bound for arbitrary refinements.

Let $O = [E_0 \cup E_1]$. It is nonempty and clopen, and safety multiplicity is exactly $1 + \mathbf{1}_O$. For a full-support probability ν , either some branch fails the bounded-density prerequisite, or all densities exist. In the latter case the images of controls 0 and 1 lie, respectively, in $[D \setminus A]$ and $[A]$; their supports are disjoint. The largest two densities consequently sum to all three almost everywhere. Their integral is $\sum_u \nu(K_u) = 1 + \nu(O) > 1$. Thus $\Theta_3(\nu) > 1$. The scalar condition implies $\Theta_3 < 1$ and also fails. $\square$

For instance $1 \leq b \leq a \leq b+c$ and $c > b$ imply (C.7) for every possible $e \leq b$: $e/(b+c-e) \leq b/c < 1$, $c/(b+c) < 1$, while $s_+ \geq 1$. Every allowed overlapping relation and permutation landing pattern is covered. Constants can be uniform when these counts have fixed bounds.

For a concrete example take $D = \{0, 1, 2, 3, 4\}$, $A = \{0, 1\}$, $B = \{2\}$, $C = \{3, 4\}$, $E_0 = \{(0, 3), (1, 4)\}$, $E_1 = \{(0, 2)\}$. Let p_0 be the identity, $p_1 = (0\ 2)$, and specify p_2 by $0 \mapsto 2$, $1 \mapsto 3$, $2 \mapsto 0$, $3 \mapsto 4$, $4 \mapsto 1$. Here $s = 4/3$, $\gamma = 7/3$, $H = 23/3$, $\theta = 13/15$, and $b_n \leq (23/3)(13/15)^n$. The target densities at first letters 0, 1, 2, 3, 4 are, respectively,

$$(0, 4/5, 2/5), \quad (0, 0, 2/5), \quad (0, 0, 1), \quad (4/5, 0, 1), \quad (4/5, 0, 2/5).$$

Thus $\Theta_3 = 9/5$. In [14] the two legal successors have first letters 4 and 1. No common successor is present. The natural $A/B/C$ merging also fails to commute with the listed permutations; this observation does not exclude every possible factor.

APPENDIX D. EXACT EVEN-SAMPLING ENTROPY OF THE GREEDY FEEDBACK

We prove $h_{A_2}(s_g) = \log \rho$ for the system of Section 7.3. The proof counts words in two directions. Certain forbidden patterns bound every greedy name from above. Parsing the remaining words into three blocks produces a recurrence with growth rate ρ . For the lower bound we realize complete block strings by inverse paths of the same greedy map; we do not need every word allowed by the upper bound to occur. On $C_0 = [0, 3)$ the actual map $R = T_{s_g}^2$ is

$$R(x) = \begin{cases} 9x/4, & 0 \leq x < 4/3, \\ 9x/4 - 3, & 4/3 \leq x < 2, \\ 9x/4 - 9/2, & 2 \leq x < 3. \end{cases} \quad (\text{D.1})$$

Each branch is a composition of two steps of s_g : at the intermediate state the same threshold 2 decides between A_0 and A_1 . Let 0, 1 denote the even-time labels A_0, A_1 .

If a core even name begins 11, its second state is in $[2, 9/4)$, its third in $[0, 9/16)$, and the next two images are in $[0, 81/64)$ and $[0, 729/256)$. The first of these is below $4/3$. If the last image is at least 2, its next image is smaller than $1953/1024 < 2$. Consequently every core even name avoids

$$111, \quad 1101, \quad 110011. \quad (\text{D.2})$$

All interval upper endpoints in these calculations are excluded by the core or its actual branch convention; $4/3$ and 2 both map to zero under R .

Every word avoiding (D.2) has a unique left-to-right parsing into the prefix-code blocks $a = 0$, $b = 10$, $c = 1100$, with no adjacent cc , followed by one of $\emptyset, 1, 11, 110$. The first two

exclusions bound one-runs and force two zeros after a full double-one run; the third prevents two complete c blocks. Let H_m count complete block strings of bit length m with no cc , with $H_m = 0$ for $m < 0$. If X and Z count strings ending in a/b and in c , respectively, their formal generating series satisfy

$$X = (z + z^2)H, \quad Z = z^4(1 + X), \quad H = 1 + X + Z.$$

Thus

$$H(z) = \frac{1 + z^4}{1 - z - z^2 - z^5 - z^6}, \quad H_m = H_{m-1} + H_{m-2} + H_{m-5} + H_{m-6} \quad (m \geq 5), \quad (\text{D.3})$$

with $H_0, \dots, H_4 = 1, 1, 2, 3, 6$ and $H_5 = 10$. The equation for ρ in (7.2) is equivalent to $\rho^{-1} + \rho^{-2} + \rho^{-5} + \rho^{-6} = 1$, whose left side is strictly decreasing. Its unique root obeys $5/3 < \rho < 7/4$. The initial counts are at most ρ^m , and the recurrence proves $H_m \leq \rho^m$ at all lengths. The incomplete-tail count is at most $H_m + H_{m-1} + H_{m-2} + H_{m-3} \leq D\rho^m$, where $D = 1 + \rho^{-1} + \rho^{-2} + \rho^{-3}$.

For a real full-space even word, either all labels are 1, or its first 0 occurs at some position j . At that moment the state is below 2 and lies in the core; its remaining word is governed by (D.2). This includes arbitrarily long waiting outside the core and the fixed endpoint 6. Therefore

$$N_n(s_g) \leq \sum_{m=0}^n D\rho^m \leq \frac{D\rho}{\rho - 1}\rho^n. \quad (\text{D.4})$$

Inverse paths furnishing the matching lower bound. The three blocks are $a = 0$, $b = 10$ and $c = 1100$. To realize their concatenations, the set of admissible endpoints must change after a c block: a second c is forbidden. We use two sets C, D_0 to keep track of which inverse path is available. They are subsets of the original state interval, not extra coordinates of the feedback.

Put $\eta = 4/9$, $L = 162/65$, $d_0 = 729/256$, and define

$$\begin{aligned} P_0 &= [0, 16/27] \cup [72/65, 4/3], & P_1 &= [L, 3), \\ C &= P_0 \cup P_1, & D_0 &= P_0 \cup [2, 550/243] \cup [L, d_0). \end{aligned}$$

Here η is the inverse slope of R . The endpoint L satisfies $L = 2 + \eta^2 L$, so the 1000 inverse path preserves the lower endpoint of P_1 . Applying the a inverse to P_1 gives $[\eta L, 4/3] = [72/65, 4/3]$, explaining the second component of P_0 . The upper cutoff d_0 is chosen so that $26/9 + \eta^4 d_0 = 3$: the c inverse must stay below the excluded endpoint of the core. The checks below verify all other endpoints.

The table gives a source for each target y . Its last column records all one-step controls, including those between the observed even times. For $b = 10$ there are two inverse paths, with control words 1000 and 1001. The first works for every target in C , but its source need not lie in C . When the source is required to lie there, we use 1000 for targets in P_1 and 1001 for targets in P_0 .

Even block	Target y	Source state	One-step controls
$a = 0$	$y \in C$	ηy	00
$b = 10$	$y \in C$	$2 + \eta^2 y$	1000
$b = 10$	$y \in P_0$	$70/27 + \eta^2 y$	1001
$c = 1100$	$y \in D_0$	$26/9 + \eta^4 y$	10100000

The 1000 path has its source in D_0 for every $y \in C$, and in $C \cap D_0$ when $y \in P_1$. The 1001 path supplies a source in C for every target in P_0 .

We check the path and its source set in each row. For a , the source ηy is in P_0 , the intermediate state is $2y/3 < 2$, and the endpoint is y ; both controls are zero. For the 1000 path, the states after its source are $8y/27$, ηy , $2y/3$, y . If $y \in P_0$, the source is in $[2, 550/243]$; if $y \in P_1$, it is in $[L, 70/27]$. Each source is at least 2, while the three subsequent control states are below 2. This gives 1000 with the stated source memberships.

For the 1001 path the states are $8/9 + 8y/27$, $4/3 + \eta y$, $2 + 2y/3$, y . The source belongs to $[70/27, 694/243) \subset P_1$. The first two subsequent control states are below 2. The last control state $2 + 2y/3$ lies in $[2, 26/9)$. Thus it uses control 1, proving 1001 at all included endpoints.

For c , its source is in $[26/9, 3) \subset P_1$, since $26/9 + \eta^4 d_0 = 3$ and d_0 is excluded from D_0 . Its successive states are

$$\begin{aligned} 4/3 + (3/2)\eta^4 y &< 3/2, & 2 + \eta^3 y &< 9/4, & (3/2)\eta^3 y &< 3/8, & \eta^2 y &< 9/16, \\ (3/2)\eta^2 y &< 27/32, & \eta y &< 81/64, & (3/2)\eta y &< 243/128 < 2, & y. \end{aligned}$$

Only its source and its second subsequent state are at least 2; this proves 10100000 with every intermediate state safe. The table now gives the relations

$$C \xrightarrow{a,b} C, \quad C \xrightarrow{c} D_0, \quad D_0 \xrightarrow{a,b} C.$$

Here an arrow $S \xrightarrow{w} T$ means that *every* endpoint in T has a source in S whose greedy path follows block w . It is this inverse coverage of the target, not an arbitrary forward choice at each source, that lets us concatenate the paths.

Given a complete block string without cc , label its initial boundary C , each boundary after c by D_0 , and each boundary after a or b by C . All successive labels then follow one of the displayed arrows. Choose the final state to be 0, which belongs to both sets, and construct the earlier states backwards. For a b step starting in D_0 , use the 1000 path. For one starting in C , use 1000 on target set P_1 and 1001 on P_0 . The a and c inverses are already specified in the table. Each backward step lands in the set required by the preceding arrow.

Running the greedy rule forwards from the resulting initial state therefore produces the prescribed even word, reaches zero, and continues safely forever. Distinct complete block strings give distinct even words because $\{0, 10, 1100\}$ is a prefix code. Hence $N_m(s_g) \geq H_m$.

Let $c_* = \min_{0 \leq j \leq 5} H_j / \rho^j > 0$. The full recurrence for $m \geq 6$ proves $H_m \geq c_* \rho^m$. This and (D.4) yield

$$h_{A_2}(s_g) = \log \rho. \quad (\text{D.5})$$

APPENDIX E. ARITHMETIC SAMPLING AND FINITE-BLOCK OPTIMIZATION

For a fixed globally safe Borel rule and arithmetic sampling $(0, p, 2p, \dots)$, a length- $n + m$ sampled pattern is determined by its first n entries and the length- m pattern starting from $T_s^{pn} x \in Q$. Consequently $N_{n+m}(s) \leq N_n(s)N_m(s)$. The logarithmic rate is a limit and equals $\inf_n n^{-1} \log N_n(s)$ (with division by τ when the control period is τ). This uses forward safety, not surjectivity or unrestricted concatenation.

For the full four-control class at A_2 , let K_n be the least actual count $N_n(s)$ among unrefined legal Borel feedbacks. It exists as a least member of a nonempty subset of $\{1, \dots, 4^n\}$. Merging refinements cannot increase counts, and unrefined rules are themselves allowed, so

$$\ell_2 = \inf_s \inf_n \frac{\log N_n(s)}{n} = \inf_n \frac{\log K_n}{n}. \quad (\text{E.1})$$

The equality exchanges two infima and yields an asymptotic upper bound from any globally implemented finite-block feedback. The minimizing feedback may depend on n , so it gives no asymptotic minimizer. Applying the formula requires actual feedback fibers, rather than an open-loop word table.

E.1. Weighted actual pattern pairs. The estimate $D_{2n} \leq N_n^2$ discards two kinds of information: some patterns use the faster B branches, and some even-odd pairs never occur on one trajectory. We retain both. The feedback in this subsection is an unrefined Borel rule for the original four-branch system; all counts refer to its physical labels. Set $a = 3/2$, $b = 5/3$, and $r_1 = a/b = 9/10$. Let

$$V_n = \{(s(T_s^{2j}x))_{0 \leq j < n} : x \in Q\}, \quad N_n = |V_n|,$$

and let I_n consist of the pairs (α, γ) of even and odd patterns from the *same* initial state. Since $T_s x \in Q$, one has $I_n \subset V_n \times V_n$; surjectivity of T_s is unnecessary. Let $B(\alpha)$ count the $B0, B1$ labels of α , and define

$$W_n = \sum_{\alpha \in V_n} r_1^{B(\alpha)}, \quad Z_n = \sum_{(\alpha, \gamma) \in I_n} r_1^{B(\alpha) + B(\gamma)}, \quad \kappa_n = Z_n / W_n^2.$$

A B label contributes the smaller inverse slope, hence the factor r_1 . The ratio κ_n measures the weight of realized pairs relative to all pairs of realized patterns. Thus $0 < \kappa_n \leq 1$ and $W_n \leq N_n$.

Proposition E.1. *With the preceding definitions, put*

$$\theta_n = -\frac{1}{n} \log \frac{W_n}{N_n}, \quad d_n = -\frac{1}{2n} \log \kappa_n.$$

Then $\theta_n, d_n \geq 0$ and

$$\frac{\log N_n}{n} \geq \log a + \theta_n + d_n. \quad (\text{E.2})$$

More precisely, if Λ_w is the slope of the complete affine word w and

$$\Omega_m = \sum_{w \text{ actual, } |w|=m} \Lambda_w^{-1}, \quad e_n = \frac{\log \Omega_{2n}}{2n},$$

then $\Omega_m \geq 1$, $e_n \geq 0$, and

$$\frac{\log N_n}{n} - \log a = \theta_n + d_n + e_n. \quad (\text{E.3})$$

Proof. For each actual complete word w , its Borel initial fiber E_w has diameter at most $6/\Lambda_w$, because two states following the same word separate by the factor Λ_w and their final states lie in $[0, 6]$. All these fibers partition Q . Consequently

$$1 \leq \sum_{w \text{ actual, } |w|=2n} \Lambda_w^{-1} = a^{-2n} Z_n.$$

The equality uses the bijection between complete words and their actual interlaced pairs. Nonempty zero-measure fibers are included. Substitution of $Z_n = \kappa_n W_n^2$ proves (E.2); the same substitution with $\Omega_{2n} = a^{-2n} Z_n$ proves (E.3).

Each original inverse branch carries Q into its safety domain. Write $G_w = g_{w_{m-1}} \circ \cdots \circ g_{w_0}$ for the affine composition associated with a word of length m . Thus the complete inverse interval $J_w = G_w^{-1}(Q)$ is contained in Q and has length $6/\Lambda_w$; every intermediate open-loop state is safe. The feedback fiber satisfies $E_w \subset J_w$. The intervals J_w for actual words cover Q , and, writing $\nu_m = \sum_w \mathbf{1}_{J_w}$, one obtains

$$6(\Omega_m - 1) = \int_Q (\nu_m - 1) dx = \sum_w (|J_w| - |E_w|) \geq 0.$$

Here $|E_w|$ denotes Lebesgue measure; no interval or convexity assumption is made on a feedback fiber. This also proves the claimed nonnegativity at every length. $\square$

In (E.3), θ_n is the cost of fast-branch weighting, d_n the cost of missing pairs, and e_n the excess of complete inverse coverage over unit mass. They are support-counting quantities, rather than empirical frequencies or Shannon entropies. The estimates transfer to arbitrary atom refinements by projecting to their physical control labels; the equalities themselves refer to those merged labels. Splitting an actual trajectory at time $2n$ also gives

$$N_{n+m} \leq N_n N_m, \quad W_{n+m} \leq W_n W_m, \quad Z_{n+m} \leq Z_n Z_m.$$

The weights multiply under this split, and the suffix is realized from a state in Q . The subadditivity argument at the beginning of this appendix therefore gives finite limits

$$h = \lim_n \frac{\log N_n}{n}, \quad p = \lim_n \frac{\log W_n}{n}, \quad t = \lim_n \frac{\log Z_n}{2n}, \quad \log a \leq t \leq p \leq h,$$

and the exact limiting decomposition $h - \log a = (h - p) + (p - t) + (t - \log a)$.

These formulas do not supply a uniformly positive correction. Even the uniformly strict one-step inequality $\Omega_1 \geq 6/5$ does not iterate: controls at 0 and 6 are different and each inverse slope is at least $3/5$, which proves that inequality, but the greedy feedback has $\Omega_1 = 4/3$ and

$$1 \leq \Omega_m(s_g) = a^{-m} D_m(s_g) \leq 10.$$

The last bound is the all-initial-state word estimate in Section 7.3. Hence $e_n(s_g) \rightarrow 0$. The complete inverse intervals overcount the true fibers at one step without producing an exponential overcount at later steps.

Example E.2 (Full support from a countable modification). There is a globally safe two- A Borel feedback for which $N_n = 2^n$, $I_n = V_n \times V_n$, and $\theta_n = d_n = 0$ at every length. It agrees with the greedy feedback on the full infinite trajectories of all but countably many initial states.

Proof. For a purely periodic binary sequence d , set

$$\pi(d) = 3 \sum_{j \geq 0} d_j a^{-(j+1)}.$$

This address is injective on all purely periodic sequences. Indeed, if two different sequences share a common period L and the same address, subtracting their periodic-sum formulas gives a nonzero integer polynomial P with leading coefficient ± 1 and $P(3/2) = 0$. If its degree is m , multiplication by 2^m leaves an odd leading term $\pm 3^m$ and only even remaining terms, a contradiction. Degree zero is immediate.

Let P_i be the addresses of the periodic sequences with first digit i , and $P = P_0 \cup P_1$. These sets are countable Borel sets, and $P_0 \cap P_1 = \emptyset$. Define $s_{\text{per}} = Ai$ on P_i and $s_{\text{per}} = s_g$ on $Q \setminus P$. The identity

$$a\pi(d) - 3d_0 = \pi(\sigma d) \in P \subset Q$$

proves safety on P , including 0 and 6; safety elsewhere is that of s_g . This one state rule is therefore safe on all infinite trajectories. Injectivity and induction show that its actual name from $\pi(d)$ is precisely d .

Any prescribed binary pattern on a nonempty finite window extends to a purely periodic sequence: choose a period larger than its largest coordinate and fill the other digits arbitrarily. The empty window has one pattern. Thus $\mathbf{P}_{s_{\text{per}}}(F) = 2^{|F|}$ for every finite F . Applying this to interlaced even and odd patterns gives $V_n = \{A0, A1\}^n$, $I_n = V_n^2$, $W_n = 2^n$, $Z_n = 4^n$, and the asserted zero corrections.

Finally $R = \bigcup_{j \geq 0} T_g^{-j} P$ is countable: each point has at most two one-step greedy preimages. Outside R the greedy trajectory never enters P , so induction gives identical states and controls under the two rules at every time. Nevertheless, their consecutive-time counting entropies are respectively $\log(3/2)$, by Proposition 7.3, and $\log 2$. Discarding countably many initial states would therefore change the counting question. No realization of all infinite nonperiodic binary names is needed or claimed. $\square$

This example excludes an unconditional positive lower bound for $\theta_n + d_n$ over the whole feedback class. Its own pattern count is already large, so it does not exclude an improved estimate restricted to feedbacks with small N_n , and it supplies no low-entropy feedback or strict full-class minimax gap.

APPENDIX F. THE OBSTRUCTION TO POINTWISE COMMON ATTAINMENT

Corollary 5.6 attains the rate of every member of a countable language family. For uncountable families, pointwise common attainment can fail even when the full feedback class has an exact common minimum. We first describe the pattern obstruction, then realize the relevant distinction in a system with four controls.

F.1. The combinatorial obstruction.

Example F.1 (A purely combinatorial uncountable obstruction). For every infinite $B \subset \mathbb{N}_0$, let

$$\Omega_B = \{x \in \{0, 1\}^{\mathbb{N}_0} : \text{supp}(x) \subset B\}.$$

At scale $\tau = 1$,

$$\mathbf{P}_{\Omega_B}(F) = 2^{\#(F \cap B)}, \quad h^*(\Omega_B; 1) = \log 2.$$

For $A = (a_i) \in \mathcal{S}$,

$$h_{\text{seq}}^A(\Omega_B; 1) = \log 2 \cdot \limsup_{n \rightarrow \infty} \frac{\#(B \cap \{a_1, \dots, a_n\})}{n}.$$

If $\mathbb{N}_0 \setminus \{a_i : i \geq 1\}$ is infinite, choose an infinite B in that complement. Otherwise choose $B = \{a_{k^2} : k \geq 1\}$. In either case the displayed upper density is zero. Therefore

$$\sup_{A \in \mathcal{S}} \inf_{\substack{B \subset \mathbb{N}_0 \\ B \text{ infinite}}} h_{\text{seq}}^A(\Omega_B; 1) = 0 < \log 2 = \inf_{\substack{B \subset \mathbb{N}_0 \\ B \text{ infinite}}} h^*(\Omega_B; 1). \quad (\text{F.1})$$

This proves only that minimax exchange is not a formal consequence of pointwise attainment for arbitrary uncountable pattern families. Every invariant-partition name system satisfies $\sigma(\Sigma_C) \subset \Sigma_C$ by suffix closure. In contrast, $\sigma(\Omega_B) \subset \Omega_B$ forces $k-1 \in B$ whenever $k \in B \setminus \{0\}$; an infinite such B must equal $\mathbb{N}_0$. Hence the proper sparse sets producing (F.1) cannot occur as these autonomous control name systems. The equation is not a deterministic control counterexample and yields no failure statement for the unrestricted Borel invariant-partition class.

F.2. A continuous family of sparse hereditary subshifts. We need an uncountable family of shift-invariant languages with two features. Each language must contain full binary patterns on its own sparse set of positions, but a fixed sampling sequence should see few of those positions for most parameters. We encode a parameter by a branch of the binary tree. Powers of four place its nodes so that a pair of positions determines their translation uniquely.

Write $\Theta = \{0, 1\}^{\mathbb{N}}$, with its fair Bernoulli probability ν . Let $\text{val}_2(v)$ be the integer represented by the binary word v , with value zero for the empty word. Set

$$j(v) = 2^{|v|} + \text{val}_2(v), \quad b_v = 4^{j(v)}, \quad B_\theta = \{b_{\theta|_\ell} : \ell \geq 0\}.$$

The index j lists the binary-tree nodes level by level: words of length ℓ receive indices $2^\ell, \dots, 2^{\ell+1} - 1$. Thus B_θ selects one marked position from each level along the branch θ .

In $K = \{0, 1\}^{\mathbb{N}_0}$ define

$$Y_\theta = \text{cl} \bigcup_{t \in \mathbb{Z}} \{y : \text{supp}(y) \subset (B_\theta + t) \cap \mathbb{N}_0\}.$$

We allow any subset of each translated support. Consequently these sets are hereditary: changing any collection of 1's to 0's keeps a sequence in the set. The closure retains this property.

Lemma F.2. *Each Y_θ is a nonempty hereditary subshift with $\sigma Y_\theta = Y_\theta$ and zero ordinary entropy. The map $\theta \mapsto Y_\theta$ is Hausdorff continuous and injective. Every Y_θ has coordinate maximal-pattern entropy $\log 2$. For each fixed sampling sequence $A = (a_i)$,*

$$\frac{1}{n} \log \mathbf{P}_{Y_\theta}(\{a_1, \dots, a_n\}) \longrightarrow 0 \quad \text{for } \nu\text{-almost every } \theta.$$

The exceptional set may depend on A .

Proof. Two marked positions will determine at most one translate of the support set. This makes finite languages locally constant in the parameter and limits how many positions of a fixed sampling window can belong to a typical support. We use these two consequences in turn.

The positive differences of $D = \{4^j : j \geq 1\}$ have unique representations. Indeed, the 4-adic valuation of $4^j - 4^i$ with $j > i$ is i , which determines i and then j . A finite pattern appears in Y_θ exactly when its set of one-positions is contained in a translate of B_θ ; cylinders are clopen and prescribed zero-positions impose no further restriction. If there are at least two one-positions, their difference fixes the candidate nodes and the translation. The remaining one-positions either give incompatible nodes, or the appearance parameters form the cylinder determined by the deepest node. Patterns with at most one one appear for every parameter. Thus each finite prefix language is locally constant in θ , proving Hausdorff continuity. The pattern 11 on $\{0, b_v - b_\emptyset\}$ detects whether θ extends a nonempty word v , proving injectivity.

Setting coordinates to zero preserves the defining supports and then their closure. Shifting replaces t by $t-1$, while prepending zero replaces it by $t+1$; compactness supplies predecessors of limit points. This proves heredity and surjectivity. An interval of length N meets any translate of D in at most $L_N = 2 + \lceil \log_4(N+1) \rceil$ points. Consequently

$$|\mathcal{L}_N(Y_\theta)| \leq (L_N + 1)N^{L_N} \quad (N \geq 2),$$

which gives zero ordinary entropy. On any n positions in B_θ all binary patterns occur, so $p_{Y_\theta}^*(n) = 2^n$.

It remains to estimate patterns on a fixed sampling sequence. For $F_n = \{a_1, \dots, a_n\}$ put $M_\theta(F_n) = \sup_t |F_n \cap (B_\theta + t)|$. This is the largest possible number of 1's in a pattern on F_n . A bound on $M_\theta(F_n)$ will therefore bound the number of patterns by counting binary words with at most that many 1's. We first list the possible translations without fixing θ . If a translate meets F_n twice, the two positions give a difference $4^j - 4^i$. Its unique representation determines both marked positions and the translation. There are therefore at most $\binom{n}{2}$ candidate translations.

We now vary θ while keeping F_n fixed. For one candidate t , there are at most n binary-tree nodes v with $b_v + t \in F_n$. If a branch contains at least k of them, one of these nodes has depth at least $k-1$, since a branch has only one node at each depth. The set of parameters extending a particular node v has measure $2^{-|v|}$. A union bound, first over the nodes and then over the candidate translations, gives

$$\nu\{M_\theta(F_n) \geq k\} \leq \binom{n}{2} n 2^{-(k-1)} \leq n^3 2^{-(k-1)} \quad (k \geq 2).$$

These events are finite unions of parameter cylinders. With $k_n = \lceil 6 \log_2(n+1) \rceil + 2$, the probabilities are summable. Borel–Cantelli gives $M_\theta(F_n) < k_n$ eventually almost surely. Every occurring pattern then has at most k_n one-positions, and

$$P_{Y_\theta}(F_n) \leq (k_n + 1)n^{k_n} \quad (n \geq 2).$$

Its normalized logarithm tends to zero. □

F.3. Four controls and a positive full-class minimum.

Proposition F.3. *There is a compact system with four distinct continuous control maps and a controlled-invariant compact set Q such that*

$$h_{\text{inv}}(Q) = 0, \quad L_1 = M_1 = \log 2,$$

yet no sampling sequence attains $h^(\mathcal{C}; Q)$ for every legal period-one Borel invariant partition $\mathcal{C}$. This already fails for a family of four-label selectors with Hausdorff continuous, compact name languages and rate $\log 4$.*

Proof. The first control component will be forced by a Thue–Morse state; this fixes the full-class minimum at $\log 2$. A second component can record one of the sparse languages

above, raising individual maximal-pattern rates without permitting one sequence to attain them all. A register makes the two choices physically distinct controls.

Let Z be the Thue–Morse subshift of Section 6.5, and let

$$\mathcal{E} = \{(\theta, y) : y \in Y_\theta\}, \quad Q = Z \times \mathcal{E} \times \{0, 1\}, \quad X = Q \sqcup \{\dagger\}, \quad U = \{0, 1\}^2.$$

Hausdorff continuity makes $\mathcal{E}$ compact. For $x = (z, \theta, y, r)$ define

$$f(x, (a, b)) = \begin{cases} (\sigma z, \theta, \sigma y, b), & a = z_0, \\ \dagger, & a \neq z_0, \end{cases} \quad f(\dagger, u) = \dagger.$$

These maps are continuous on clopen matching regions. At each state exactly the two controls with $a = z_0$ are safe. Their successors differ in the register coordinate b , so the four control maps are distinct and Q is controlled invariant. The register is a state coordinate of this system; the feedback has no additional memory or time variable.

For every legal Borel partition, the first control label of each atom must equal z_0 . The first state coordinate always shifts, independently of every Borel mixture and atom refinement. Label projection therefore gives $P_C(F) \geq P_Z(F)$ for each finite F . The two-atom clopen feedback $(z_0, 0)$ attains equality for every F . It follows that

$$\inf_{C \in \mathfrak{C}_{\text{Bor}}(Q, 1)} h_{\text{seq}}^A(C; Q) = h_{\text{seq, coord}}^A(Z) \quad \text{for every } A, \quad M_1 = \log 2.$$

The schedule in Proposition 6.3 gives $L_1 = \log 2$. A control word of length N is safe exactly when its first components match the initial Z word. Thus $r_{\text{inv}}(N, Q) = |\mathcal{L}_N(Z)| \leq 8N$ and $h_{\text{inv}}(Q) = 0$.

For $\eta \in \Theta$, use the Borel selector

$$s_\eta(z, \theta, y, r) = (z_0, \mathbf{1}_{\{\theta=\eta\}} y_0).$$

Its four fibers are nonempty, Borel, and assigned different controls. Its actual name set is $Z \times Y_\eta$: the slice $\theta = \eta$ realizes the full product, and the other slices add only the zero second sequence. Hence

$$P_{s_\eta}(F) = P_Z(F)P_{Y_\eta}(F).$$

The binary-addition argument in Proposition 6.3 also gives all patterns on any finite subset of $\{4^j : j \geq 0\}$: extend the prescribed digits to the largest exponent and apply the same parity correction. Thus both factors realize 2^n patterns on any n positions in B_η . It follows that $p_{s_\eta}^*(n) = 4^n$ and $h^*(s_\eta) = \log 4$. Their ordinary word entropy is zero by the word bounds for Z and Y_η , and their name languages vary Hausdorff continuously with η .

Fix A . By Lemma F.2, the normalized logarithm of the second pattern factor tends to zero almost surely. Therefore

$$h_A(s_\eta) = h_{\text{seq, coord}}^A(Z) \leq \log 2 < \log 4 \quad \text{for } \nu\text{-almost every } \eta.$$

This disproves pointwise common attainment. In the designated family one also has $\max_A \inf_\eta h_A(s_\eta) = \log 2 < \min_\eta h^*(s_\eta) = \log 4$. For the full Borel class the lower-rate selector $(z_0, 0)$ is available, and the exact equality $L_1 = M_1$ remains valid. $\square$

The compact family here consists of name languages, in the Hausdorff topology; no topology or compactness is imposed on the Borel selectors. The example separates attaining every feedback's own rate from attaining the minimum over all feedbacks.

USE OF AI TOOLS

OpenAI ChatGPT/Codex assisted with mathematical exploration and candidate proofs, comparison of primary literature, and manuscript preparation, including language editing and checks of the LaTeX source and compiled PDF.

REFERENCES

- [1] Debauche, V., Della Rossa, M., & Jungers, R. M. (2022). Comparison of path-complete Lyapunov functions via template-dependent lifts. *Nonlinear Analysis: Hybrid Systems*, 46, 101237. doi:10.1016/j.nahs.2022.101237. Author version: [arXiv:2110.13474v1](https://arxiv.org/abs/2110.13474v1).
- [2] Goodman, T. N. T. (1974). Topological sequence entropy. *Proceedings of the London Mathematical Society*, 29(2), 331–350. doi:10.1112/plms/s3-29.2.331.
- [3] Huang, W., Maass, A., & Ye, X. (2004). Sequence entropy pairs and complexity pairs for a measure. *Annales de l'Institut Fourier*, 54(4), 1005–1028. doi:10.5802/aif.2041.
- [4] Huang, W., & Ye, X. (2009). Combinatorial lemmas and applications to dynamics. *Advances in Mathematics*, 220(6), 1689–1716. doi:10.1016/j.aim.2008.11.009.
- [5] Hulse, P. (1982). Sequence entropy and subsequence generators. *Journal of the London Mathematical Society*, 26(3), 441–450. doi:10.1112/jlms/s2-26.3.441.
- [6] Kamae, T., & Zamboni, L. (2002). Maximal pattern complexity for discrete systems. *Ergodic Theory and Dynamical Systems*, 22(4), 1201–1214. doi:10.1017/S0143385702000585.
- [7] Kawan, C. (2013). *Invariance entropy for deterministic control systems: An introduction*. Lecture Notes in Mathematics, Vol. 2089. Springer. doi:10.1007/978-3-319-01288-9.
- [8] Kerr, D., & Li, H. (2007). Independence in topological and C*-dynamics. *Mathematische Annalen*, 338, 869–926. doi:10.1007/s00208-007-0097-z.
- [9] Kerr, D., & Li, H. (2009). Combinatorial independence in measurable dynamics. *Journal of Functional Analysis*, 256(5), 1341–1386. doi:10.1016/j.jfa.2008.12.014.
- [10] Romagnoli, P.-P. (2007). A remark on the topological entropies of covers and partitions. *Studia Mathematica*, 182(3), 273–281. doi:10.4064/sm182-3-6.
- [11] Tomar, M. S., Kawan, C., & Zamani, M. (2022). Numerical over-approximation of invariance entropy via finite abstractions. *Systems & Control Letters*, 170, 105395. doi:10.1016/j.sysconle.2022.105395.

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